Turning points in primary care disease trajectories

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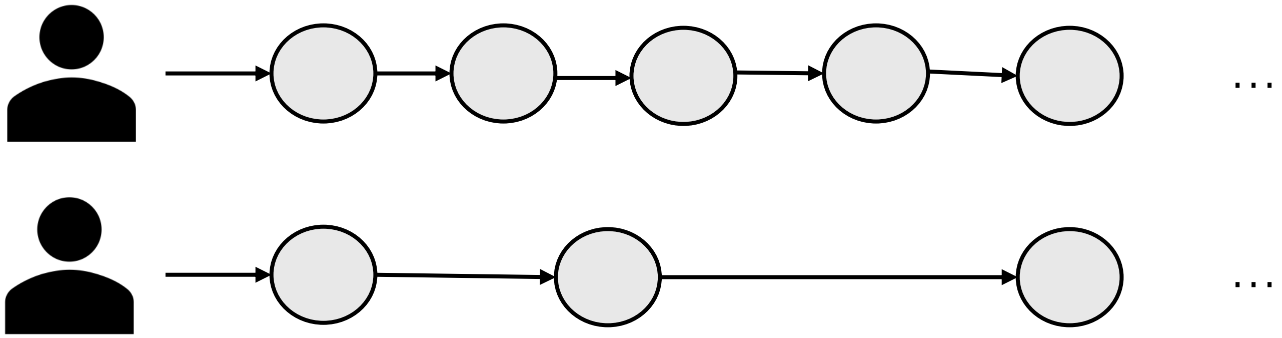
Joint work with Nathaniel Hendrix, Diego Guerrero, Laust Mortensen and David Rehkopf

Disclosure of speaker's interests

No (potential) conflict of interests
For more information see https://www.iagg2026.org/resources/uploads/sites/137/2026/01/Format-of-disclosure-slide-for-speakers-ENG.pdf

Two 55 year old women diagnosed with **diabetes**



Each person has a health **trajectory**

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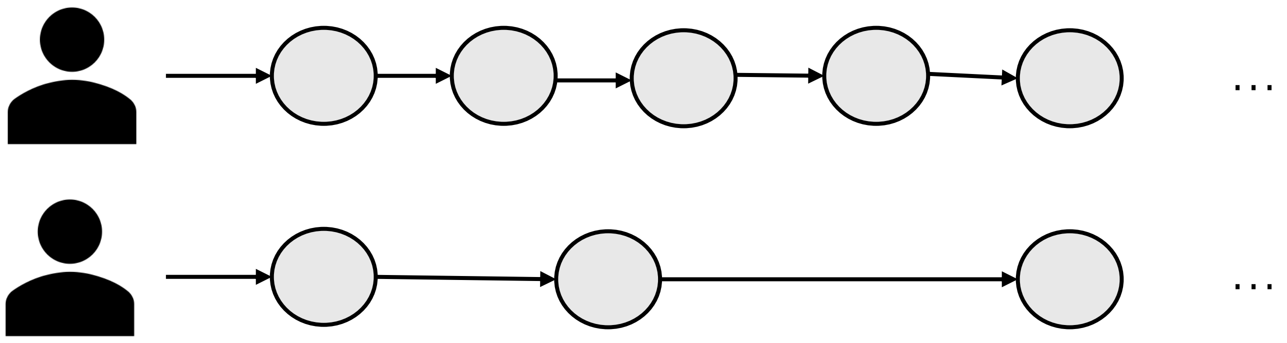

Worse outcomes

Better outcomes

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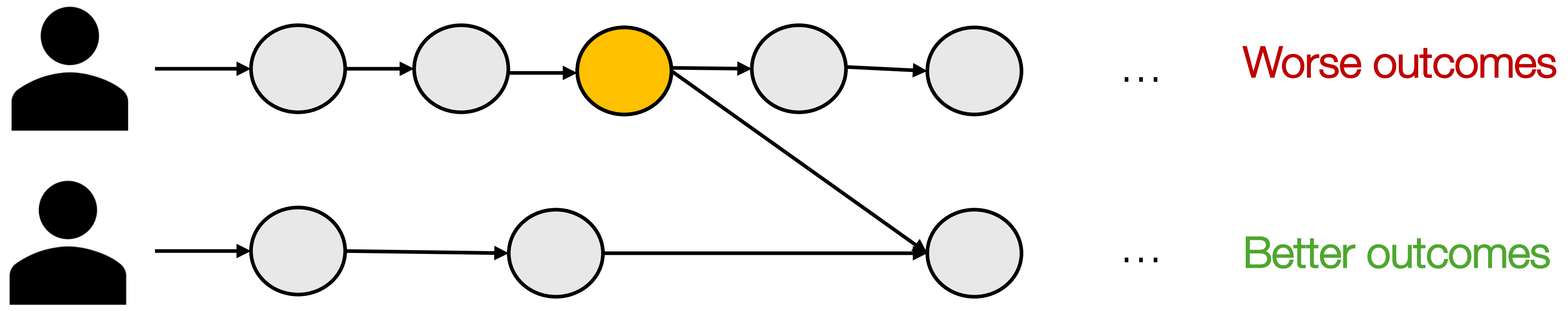

Each person has a health **trajectory**

Are there diagnoses that persistently redirect a patient's trajectory to one with better outcomes?

The idea of **"turning points"** is
theorized but hard to operationalize
But really important for intervention

Two 55 year old
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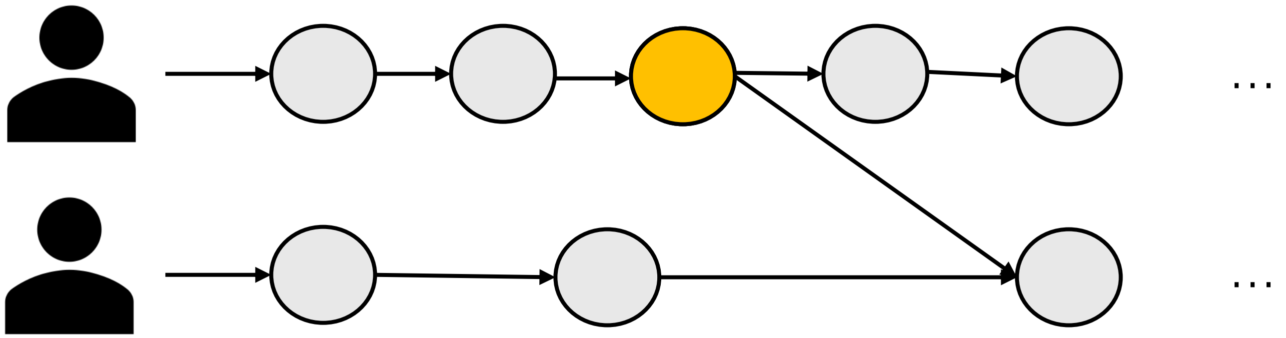
with **diabetes**



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Elder (1986). Military times and turning points in men's lives. *Developmental Psychology*.
Sampson & Laub (1992). Crime and deviance in the life course. *Annual Review of Sociology.*
Rutter (1996). Transitions and turning points in developmental psychopathology: As applied to the age span between childhood and mid-adulthood. *International Journal of Behavioral Development.*
Gotlib & Wheaton (1997). Stress and adversity over the life course: Trajectories and turning points. *Cambridge University Press.*

A lot of prior work uses European registry data

Temporal disease trajectories condensed from population-wide registry data covering 6.2 million patients

Anders Boeck Jensen[1,2], Pope L. Moseley[2,3], Tudor I. Oprea[1,3,4], Sabrina Gade Ellesøe[2], Robert Eriksson[1,2], Henriette Schmock[5], Peter Bjødstrup Jensen[2], Lars Juhl Jensen[2] & Søren Brunak[1,2]

2014

Unraveling cradle-to-grave disease trajectories from multilayer comorbidity networks

Check for updates

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2024

Disease trajectory browser for exploring temporal, population-wide disease progression patterns in 7.2 million Danish patients

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2020

Mapping multimorbidity progress among 190 diseases

Shasha Han[1,2,3], Sairan Li[1], Yunhaonan Yang[4], Lihong Liu[5], Libing Ma[6], Zhiwei Leng[7], Frances S. Mair[8], Christopher R. Butler[9,10], Bruno Pereira Nunes[11,12], J. Jaime Miranda[13,14], Weizhong Yang[1,2,3], Ruitai Shao[1,2,3] & Chen Wang[1,2,3,15]

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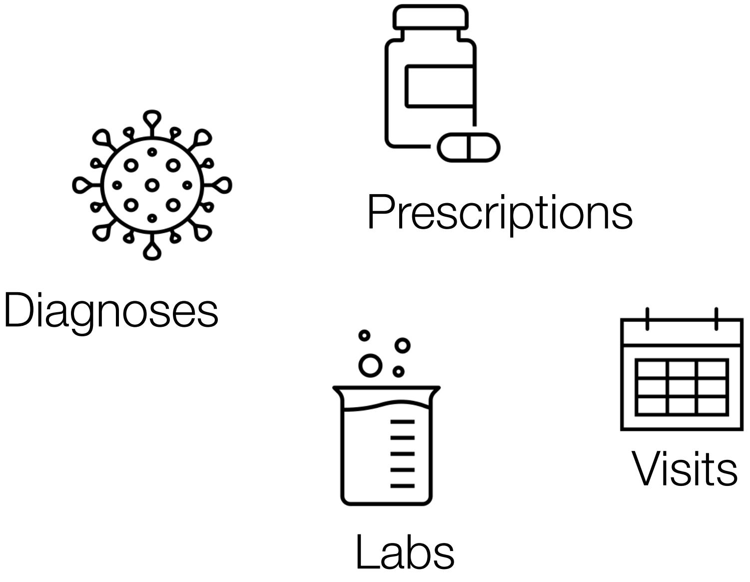
Prior work often uses **hospital data**, often from a **single care system**, or for a **single disease**. And focuses on **predicting outcomes**, without understanding where their paths separate.

Pollington et al. (2026) Evaluation of trajectory analysis for disease risk assessment: A scoping review. *JAMIA*

Primary care is uniquely rich!

American Family Cohort data

- 7.9 million patients
- 1,331 clinics in the U.S.
- 4,573 providers
- 2017-2024



This is the setting where divergence in trajectories is likely most visible

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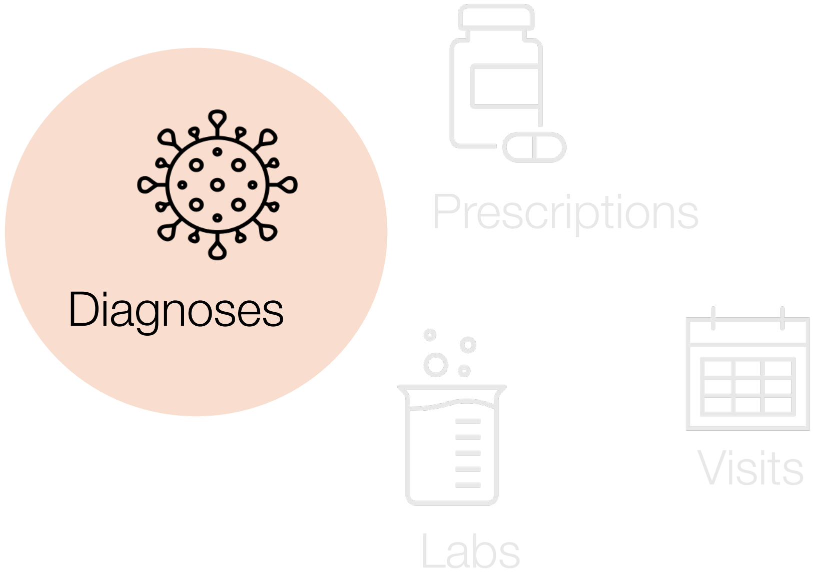

Would like to incorporate the other info eventually!

Primary care is uniquely rich!

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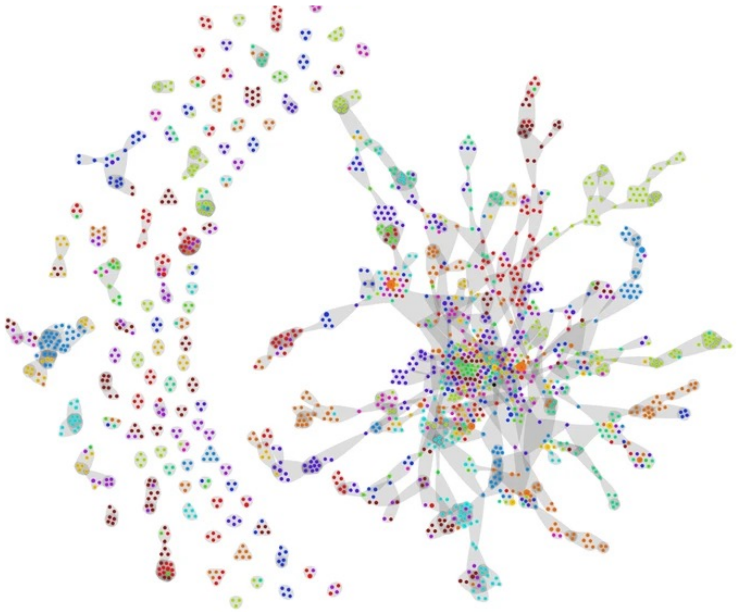
Use a **network**, a way of representing complex data

Maps how **diseases cluster and progress together**

Let’s the strongest relationships emerge



What is the disease trajectory landscape in U.S. primary care?

How can we operationalize turning points?

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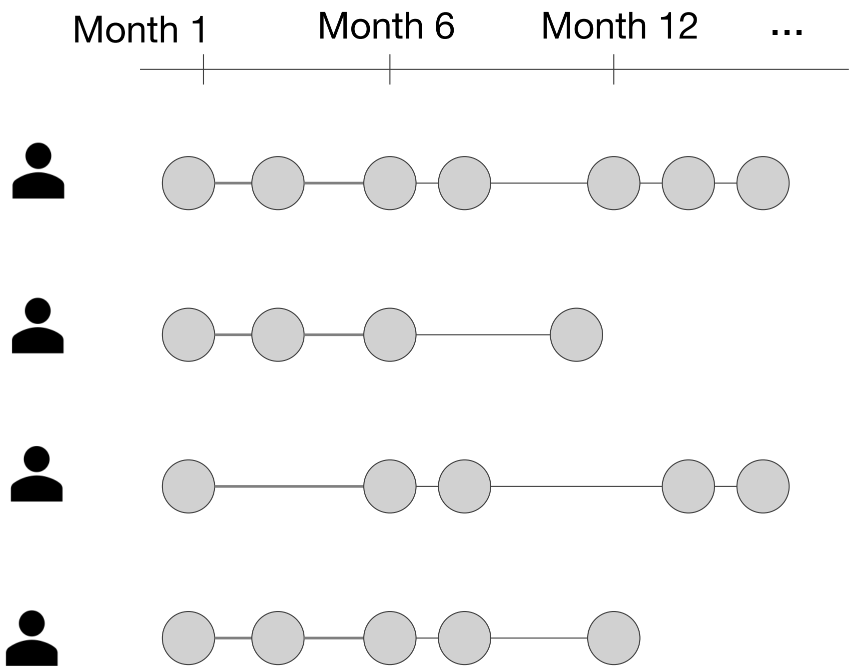
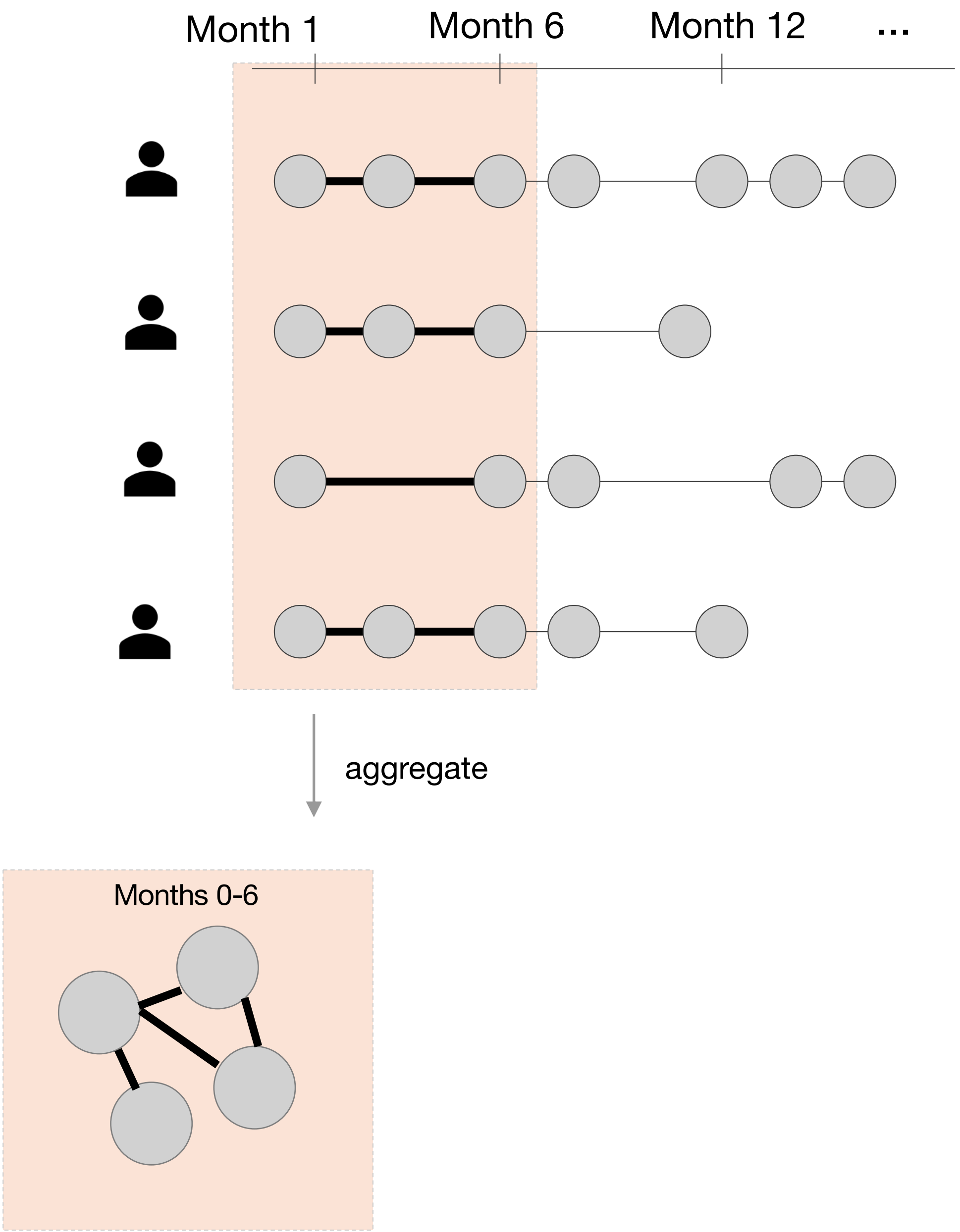


Each person has a sequence of diagnoses over 5 years



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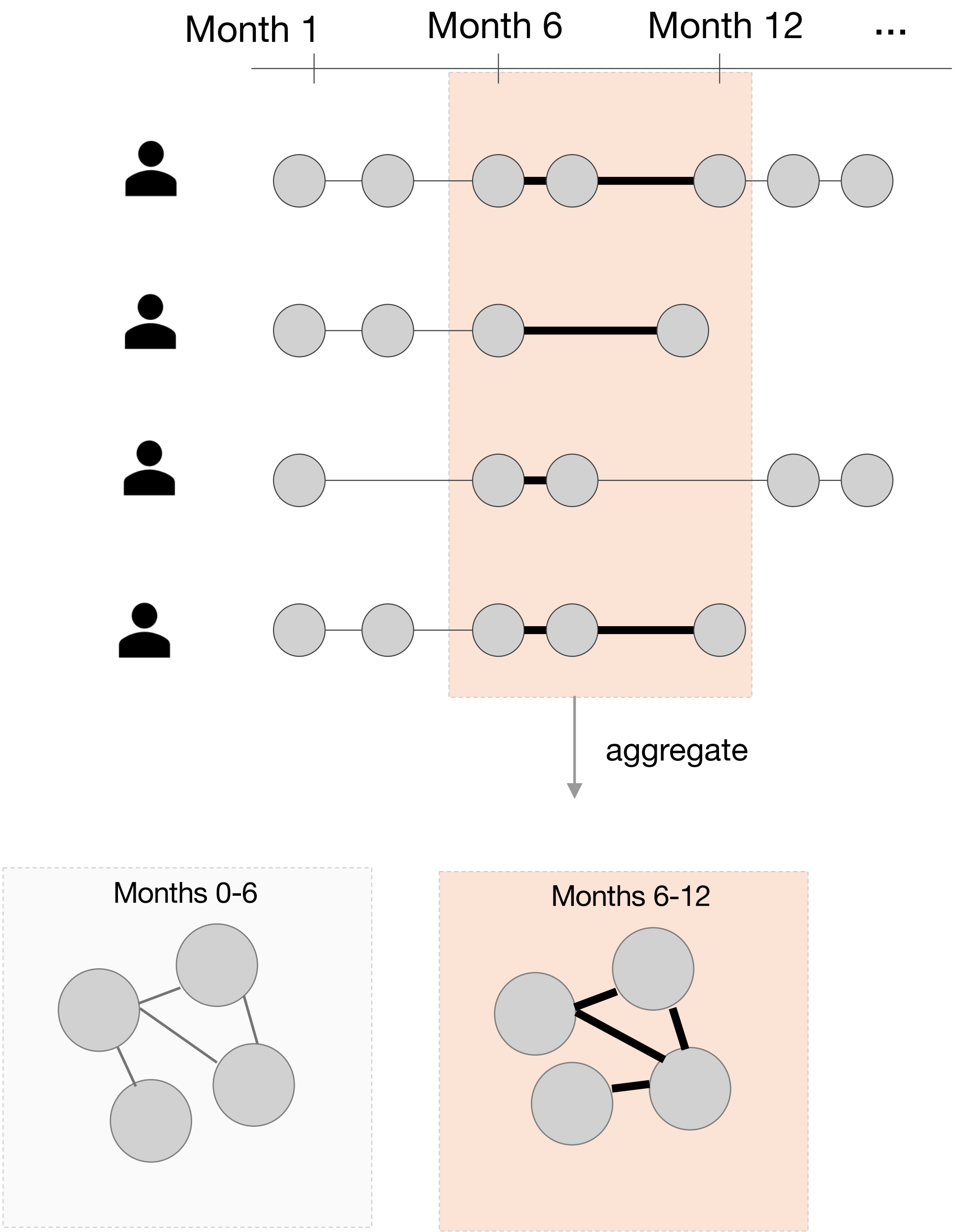
An **edge** between two diseases means that pair **co-occurs** at a greater rate than expected in the **population**

— undirected: co-occurrence within bin (RR ≥ 1.5)



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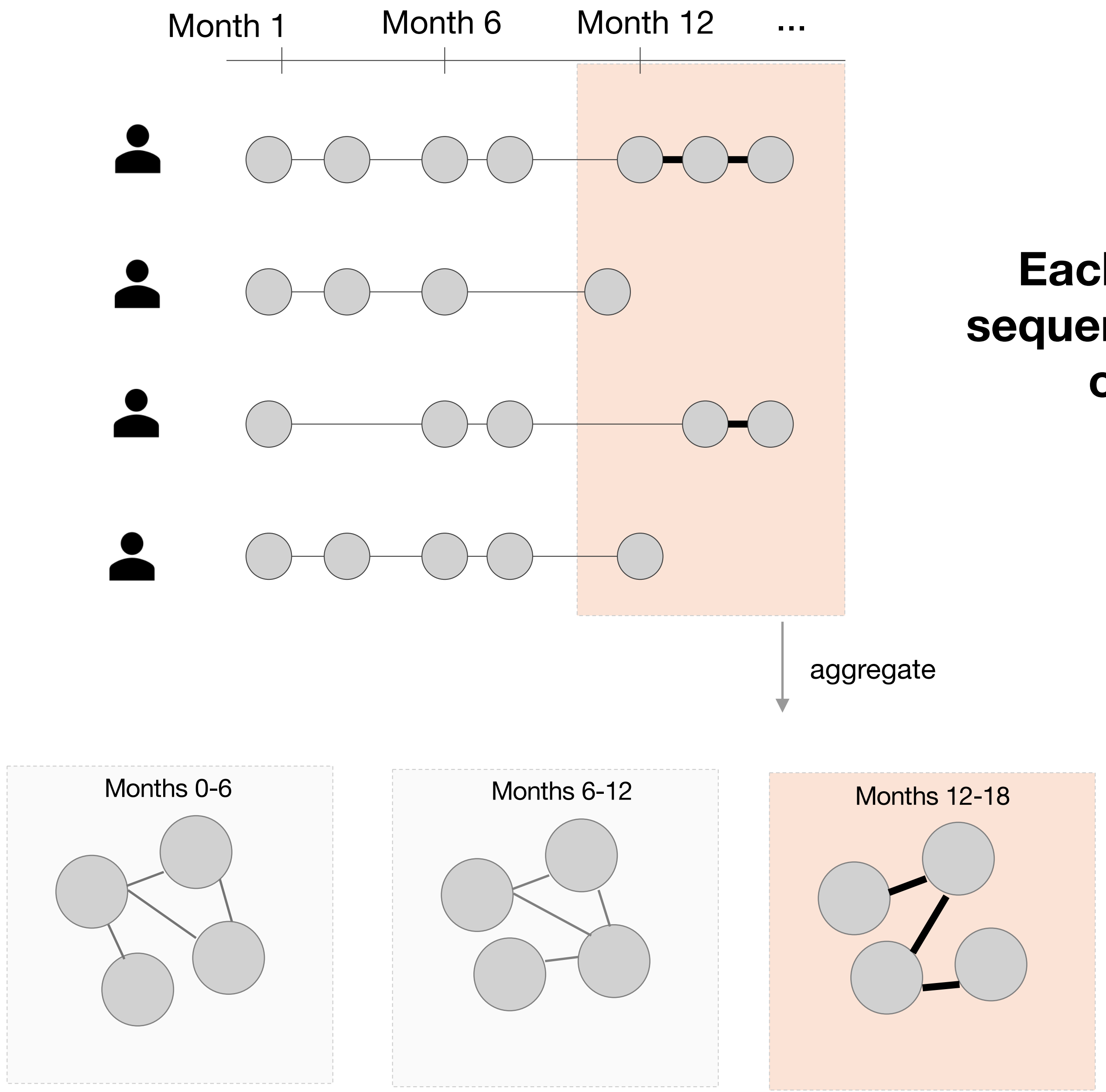

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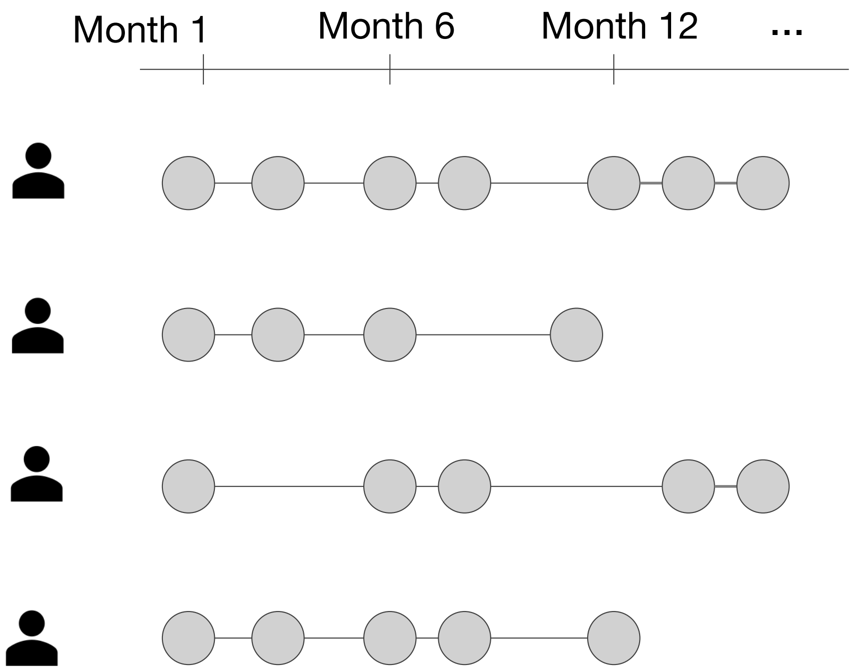

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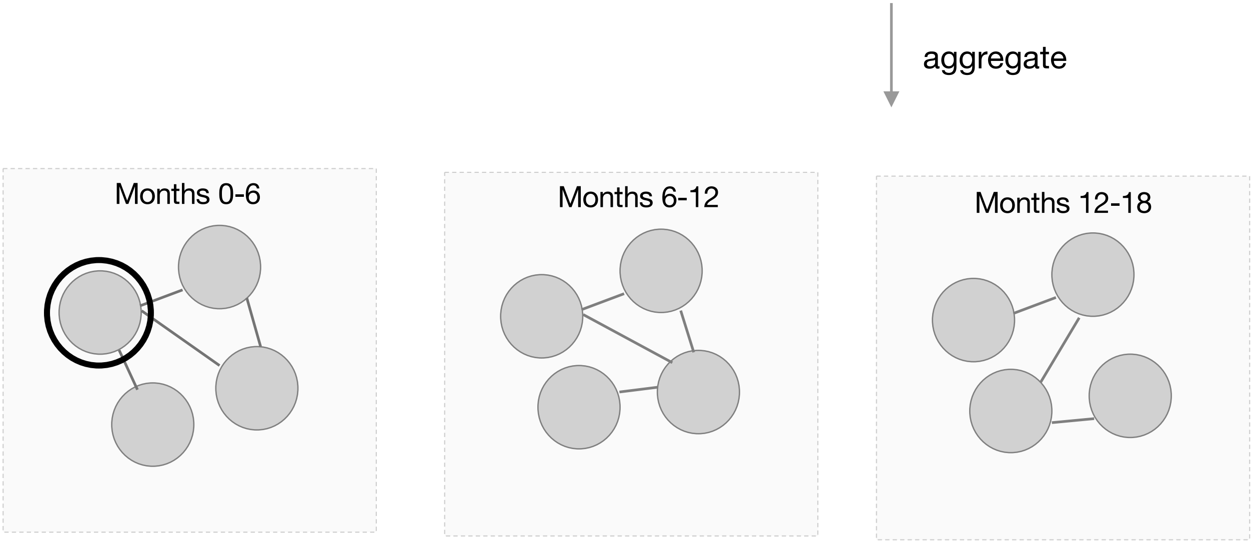
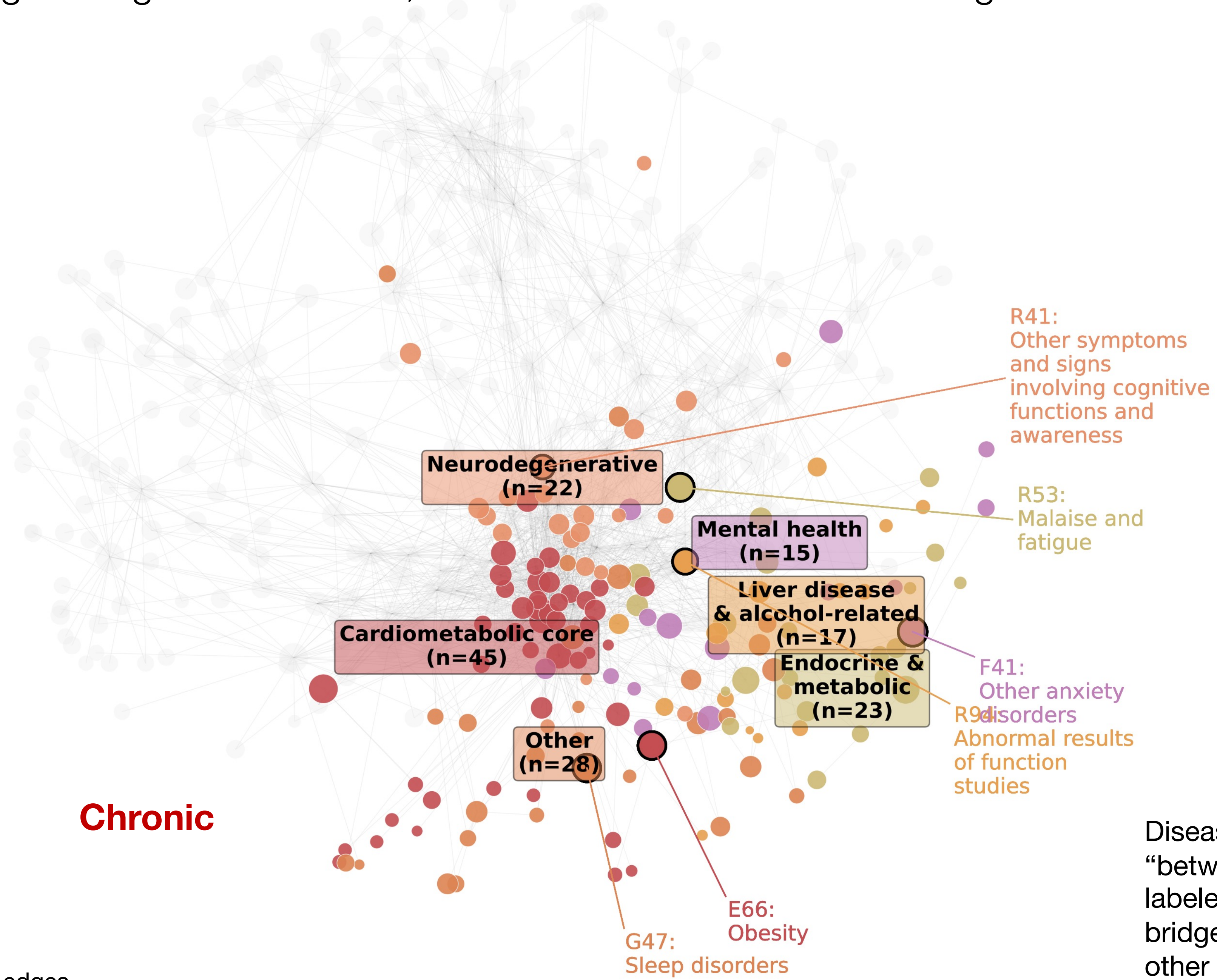


High "betweenness" centrality means that disease is a **bridge** between many other diseases

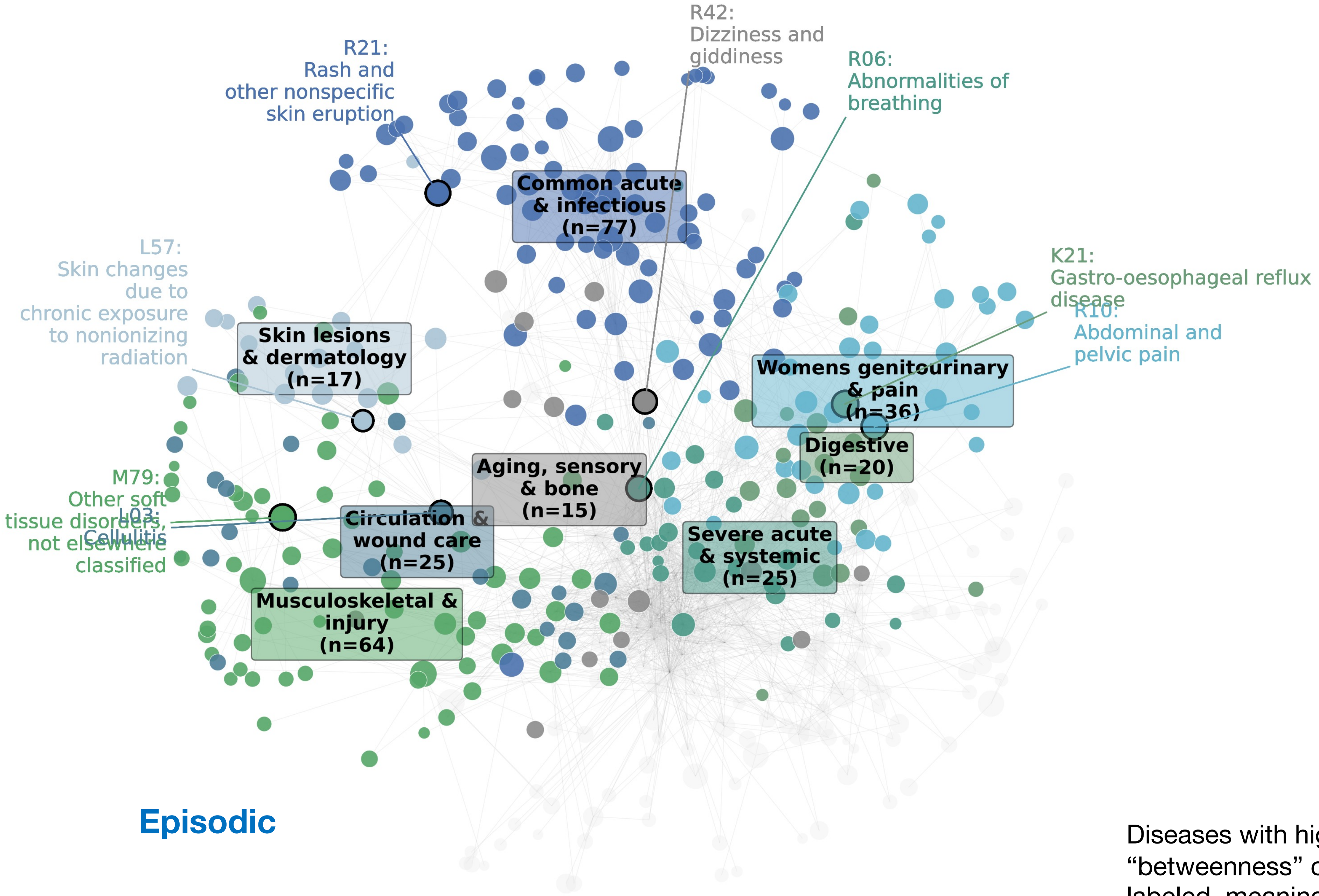
— undirected: co-occurrence within bin (RR ≥ 1.5)

When we aggregate edges across time, two main disease clusters emerge



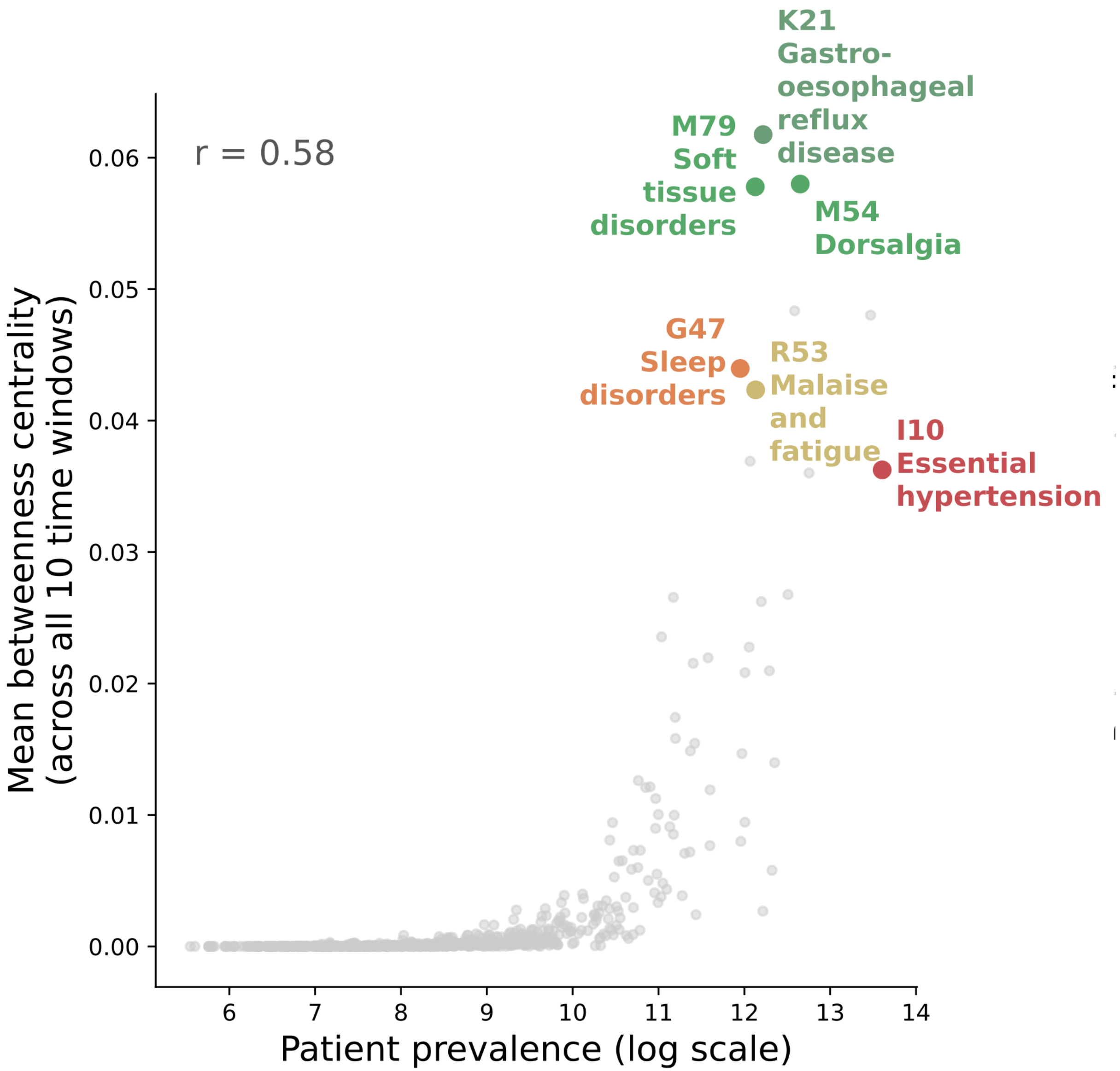
Diseases with high "betweenness" centrality are labeled, meaning they are bridge nodes between many other diseases

Node size represents number of edges



Episodic

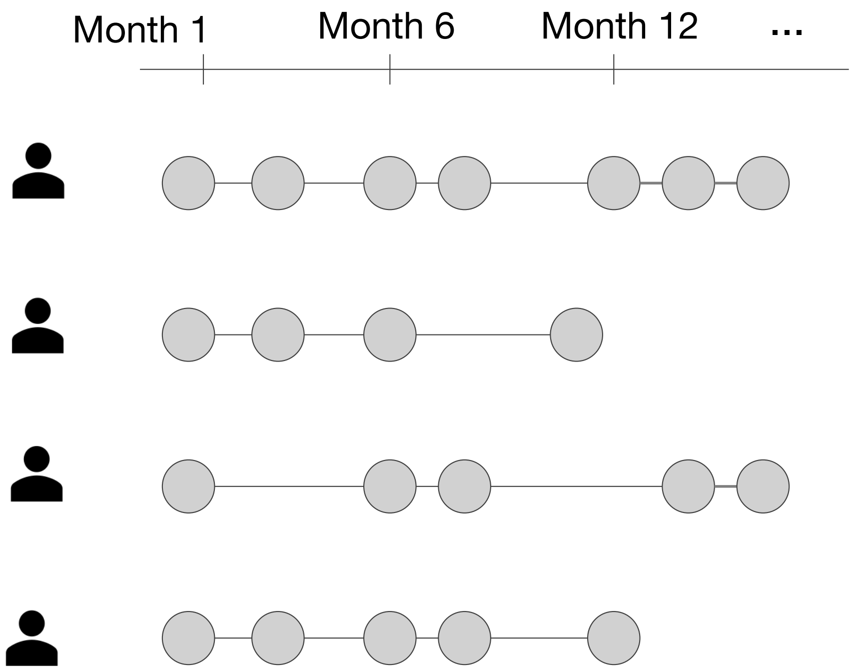
Diseases with high "betweenness" centrality are labeled, meaning they are bridge nodes between many other diseases



A small set of diseases connect groups of diseases (bridge nodes)

They tend to be prevalent but higher prevalence =/= higher betweenness

The highest ones are episodic



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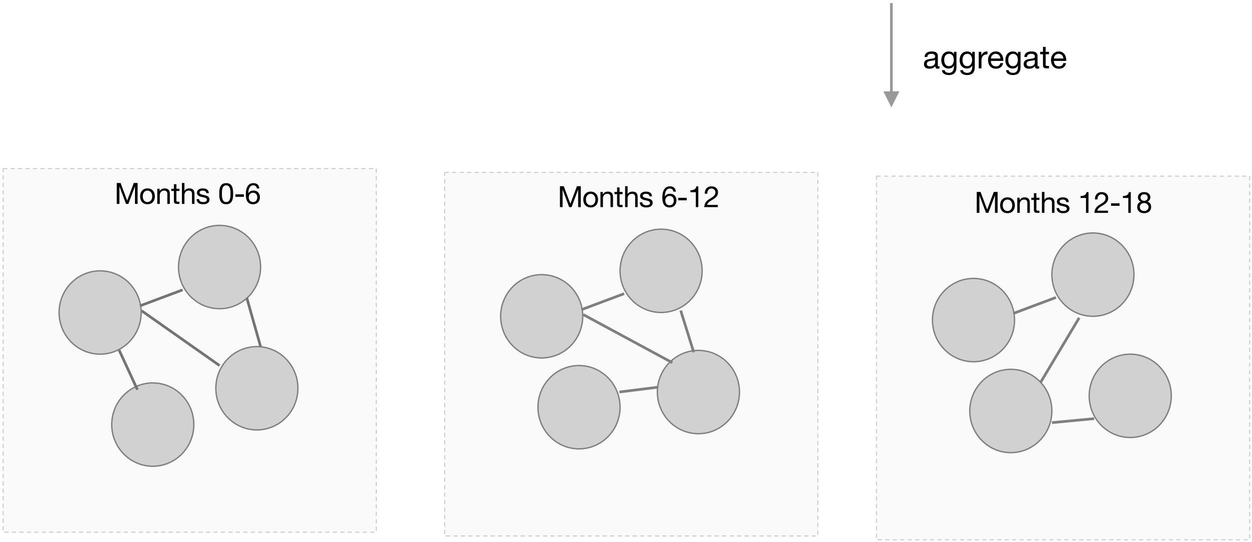
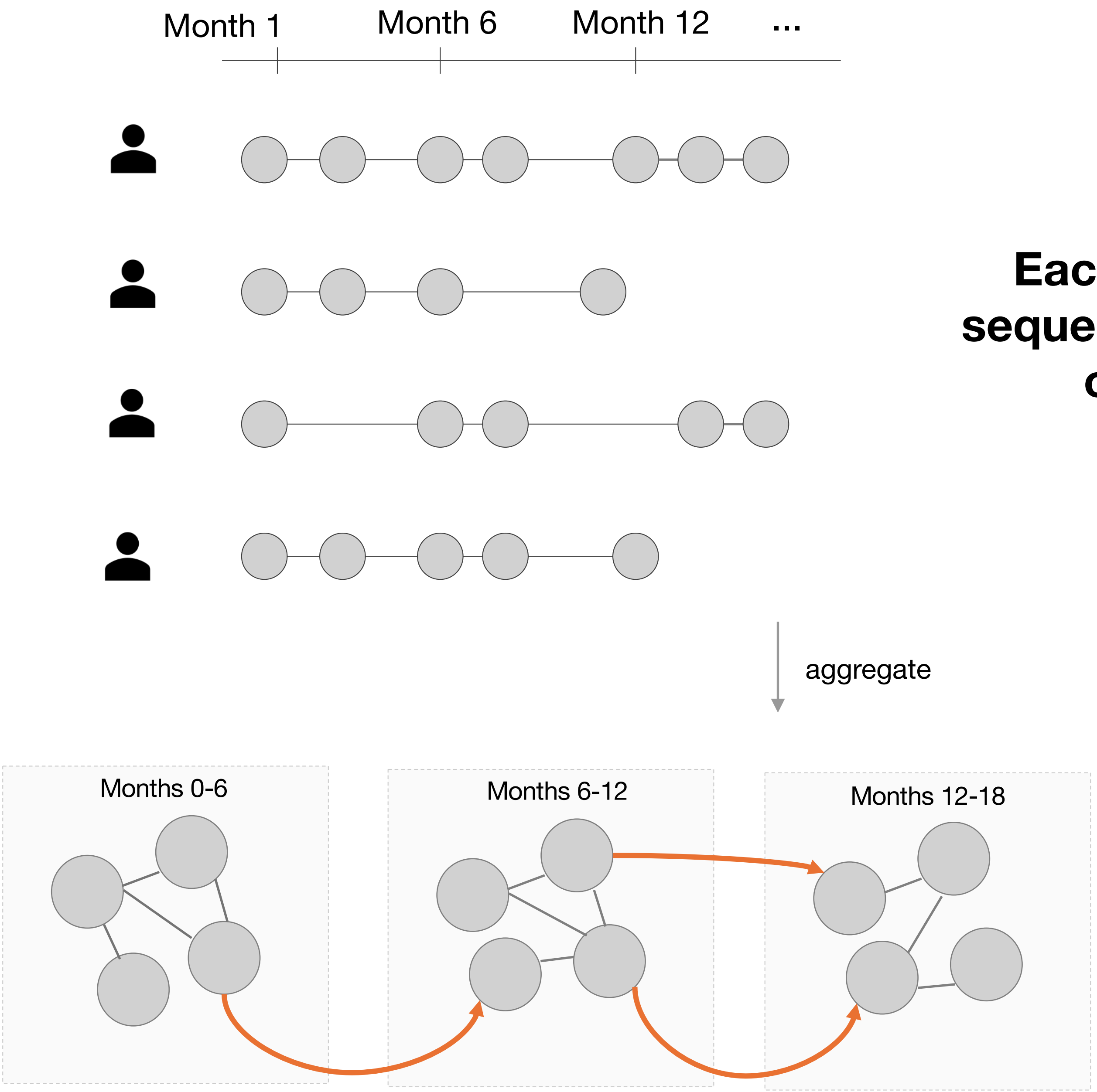


These edges tell us which diseases cluster together (comorbidity)

But the order the diseases were diagnosed matters too

— undirected: co-occurrence within bin (RR ≥ 1.5)

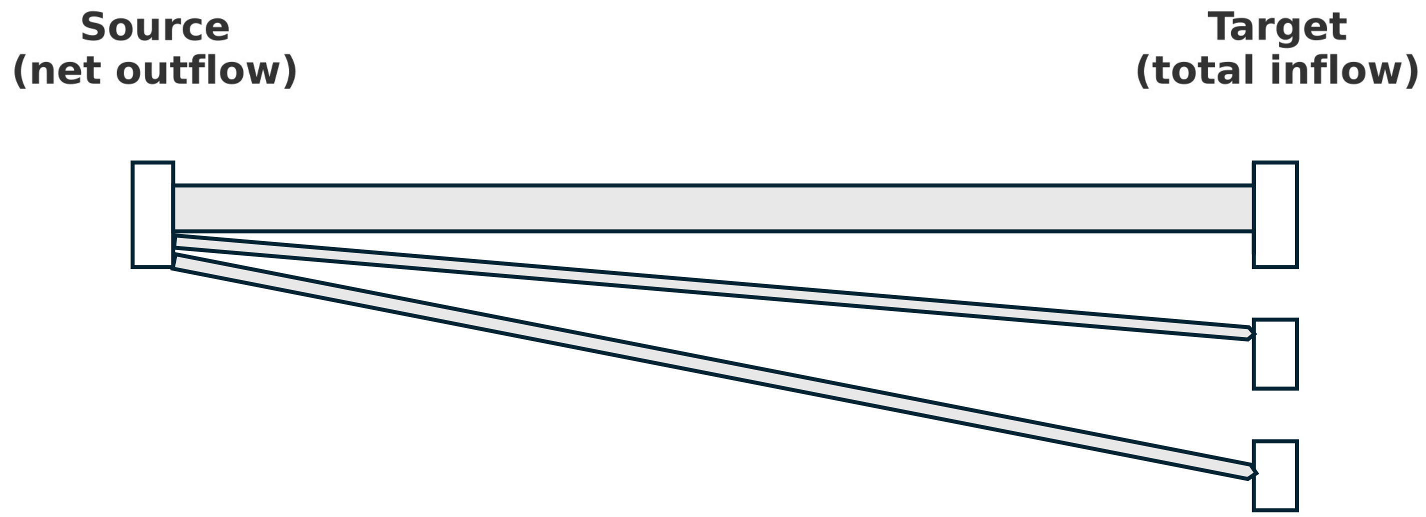


Each person has a sequence of diagnoses over 5 years

Arrows mean an earlier diagnosis predicts the later one

— undirected: co-occurrence within bin (RR ≥ 1.5)
→ directed: A in bin t predicts B in bin t+1 (RR ≥ 1.5)

Which diseases are upstream versus downstream?



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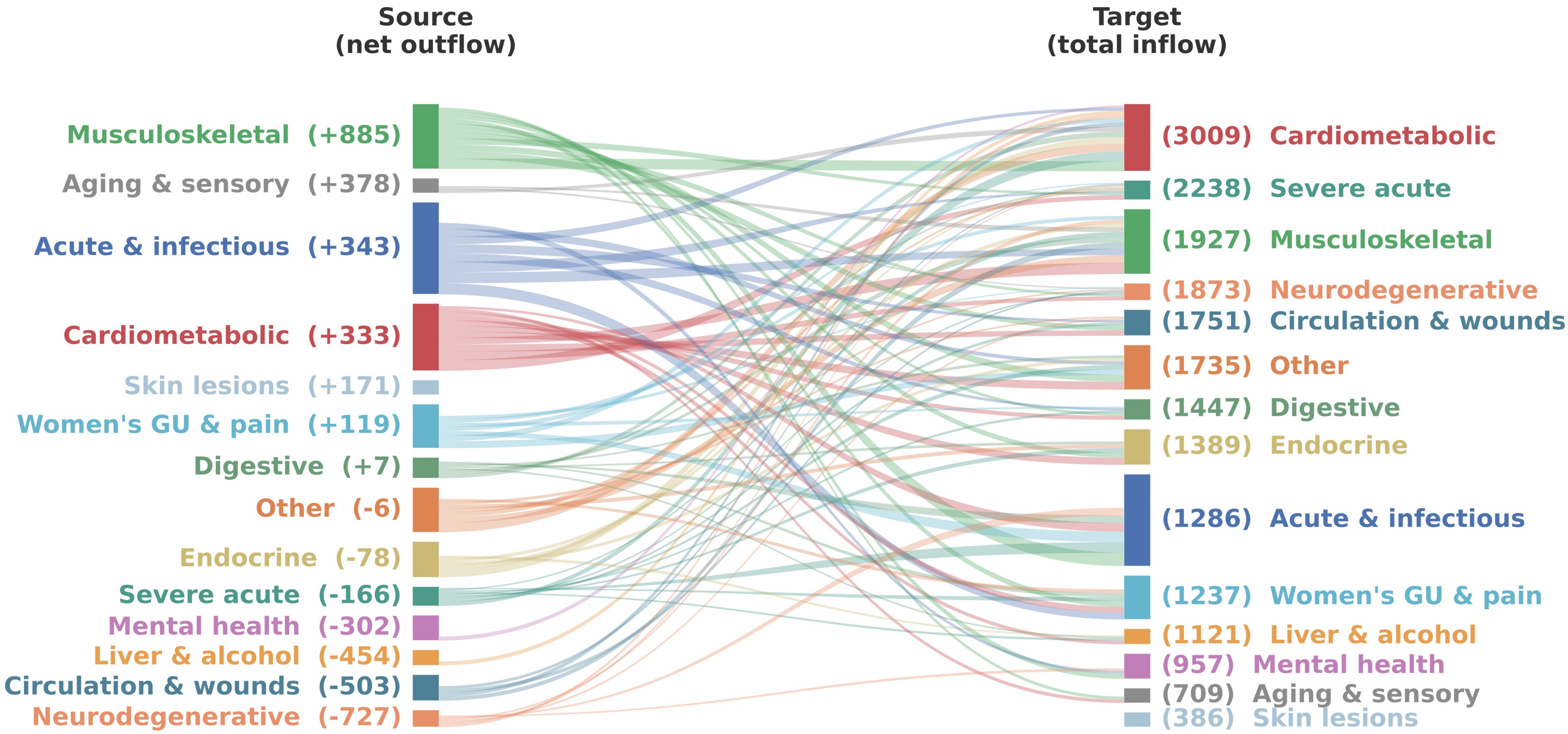

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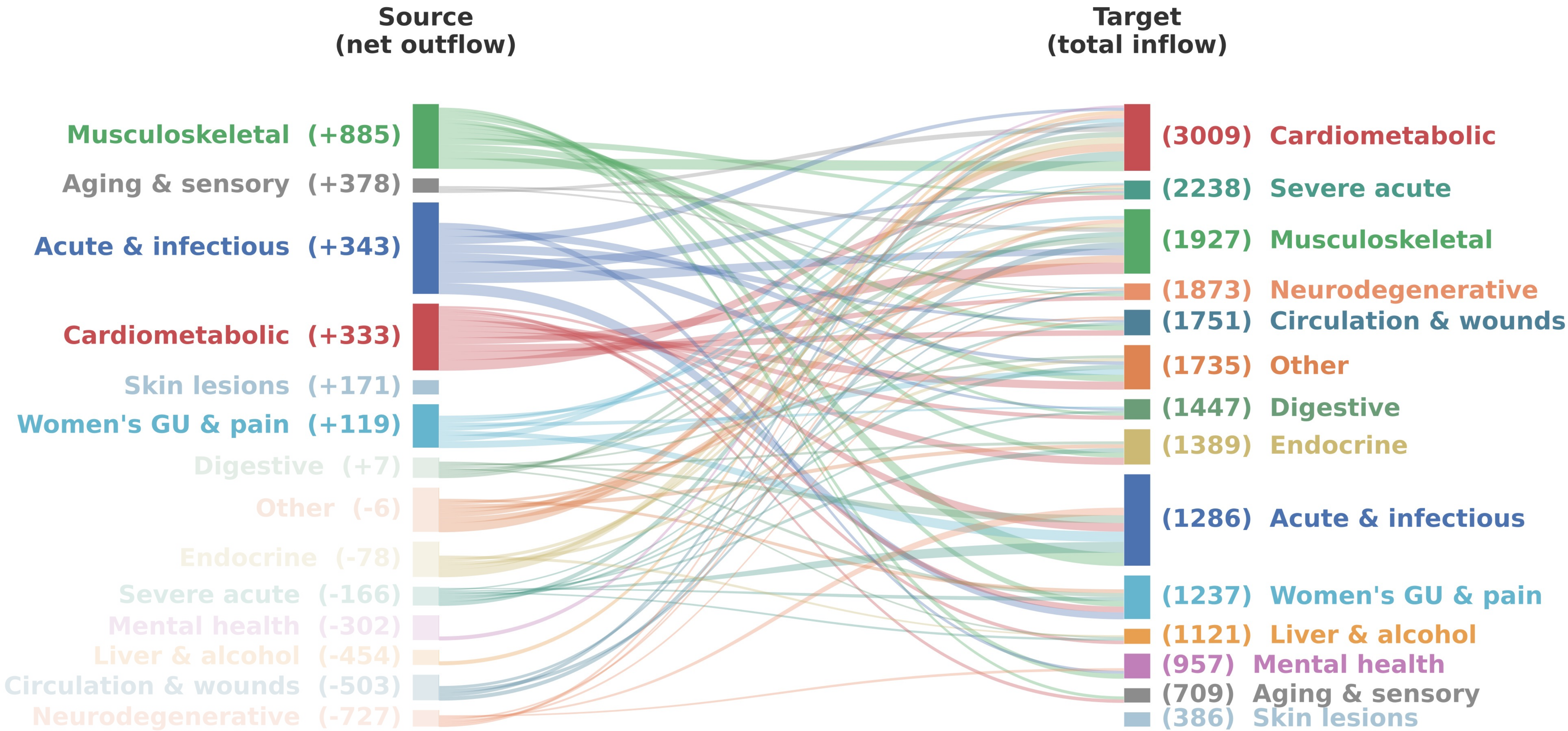

These disease clusters are **net sources**, meaning they predict more diseases upstream than they are predicted downstream

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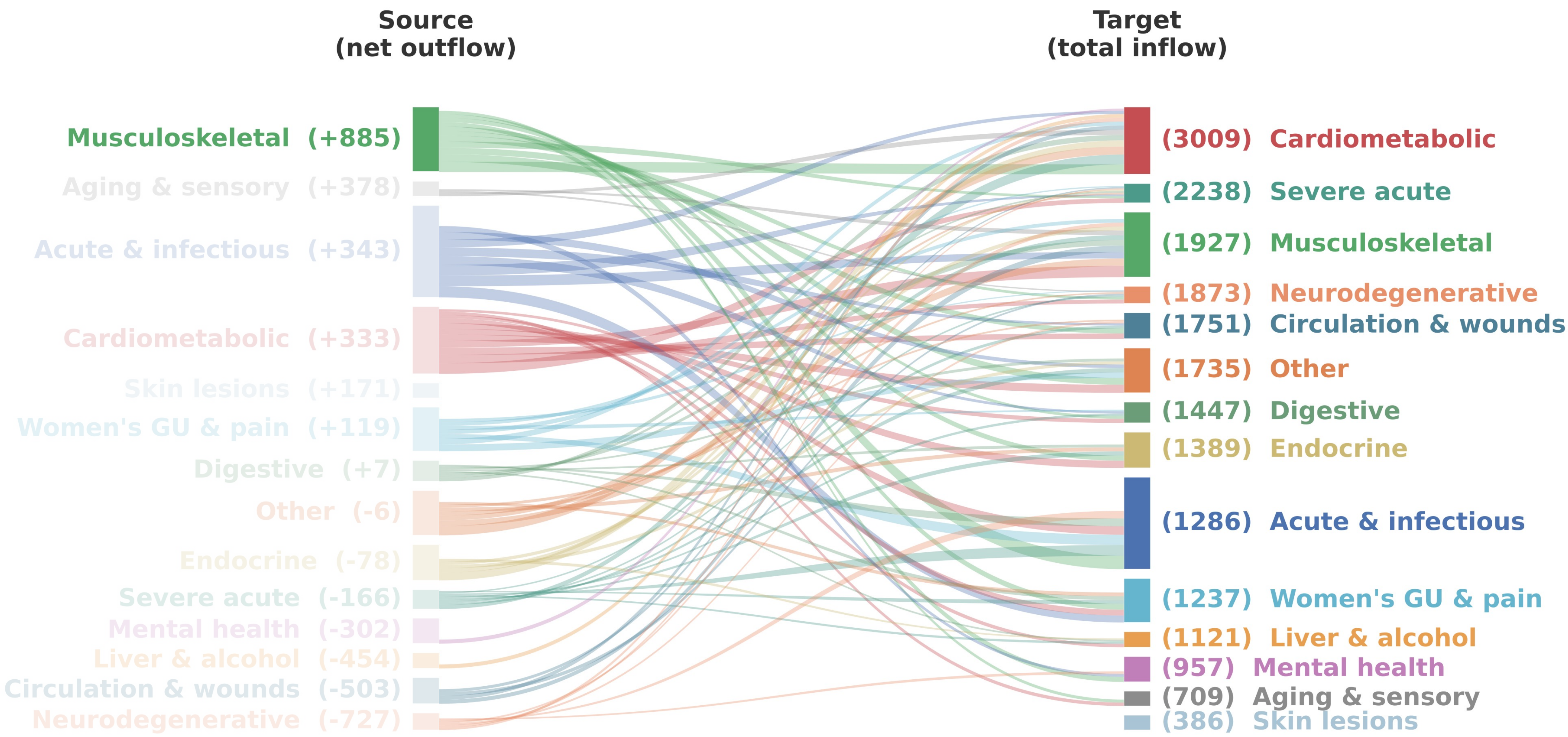

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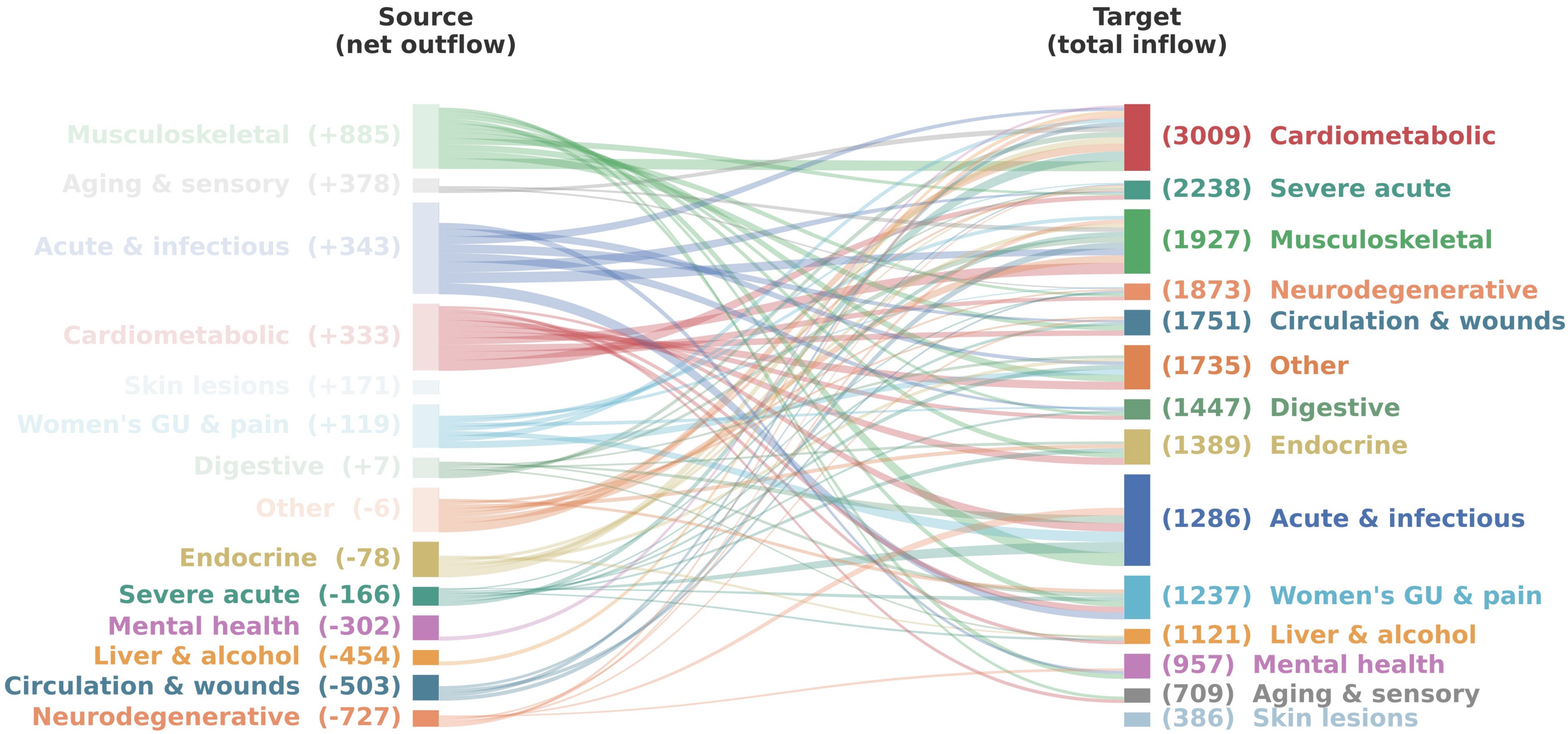


These disease clusters are **net sinks**, meaning they are more often predicted downstream

What is the disease trajectory landscape in U.S. primary care?

Population

Tells us which diseases connect the landscape, how chronic disease progresses
Doesn't tell us why our two women end up in different places

What is the disease trajectory landscape in U.S. primary care?

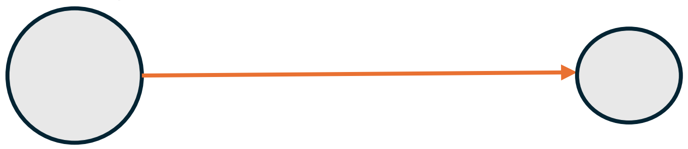
Population

↓

How can we operationalize turning points?

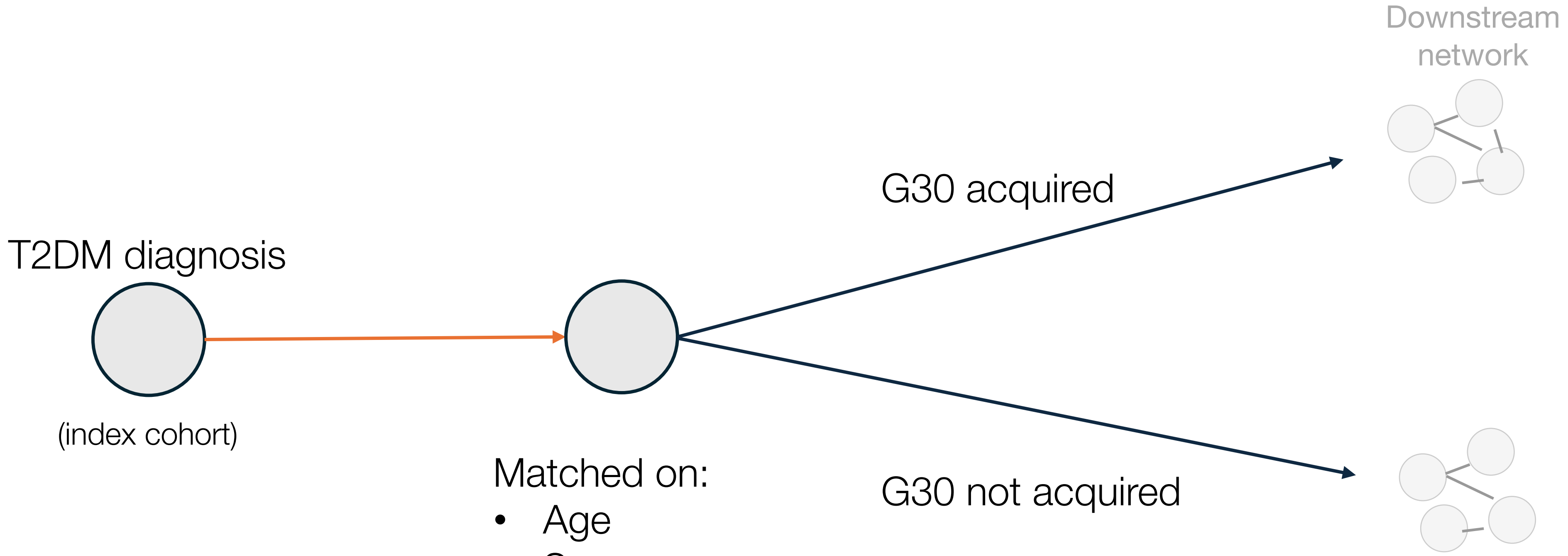
Individual

Counterfactual design



Matched on:
- Age
- Sex
- Race
- Baseline comorbidity count
- Follow-up length

Counterfactual design



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Compare:

Does acquiring this condition reorganize the downstream disease landscape?

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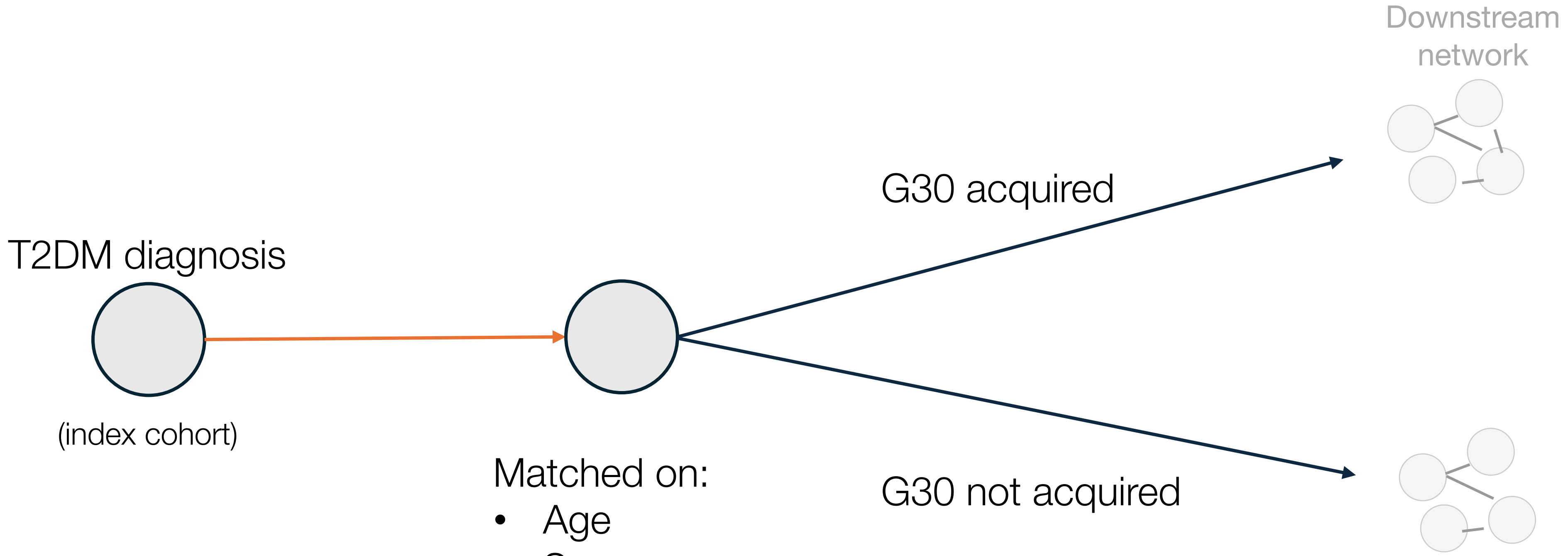

Matched on:
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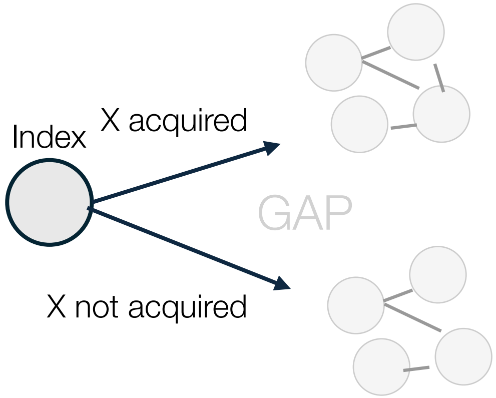
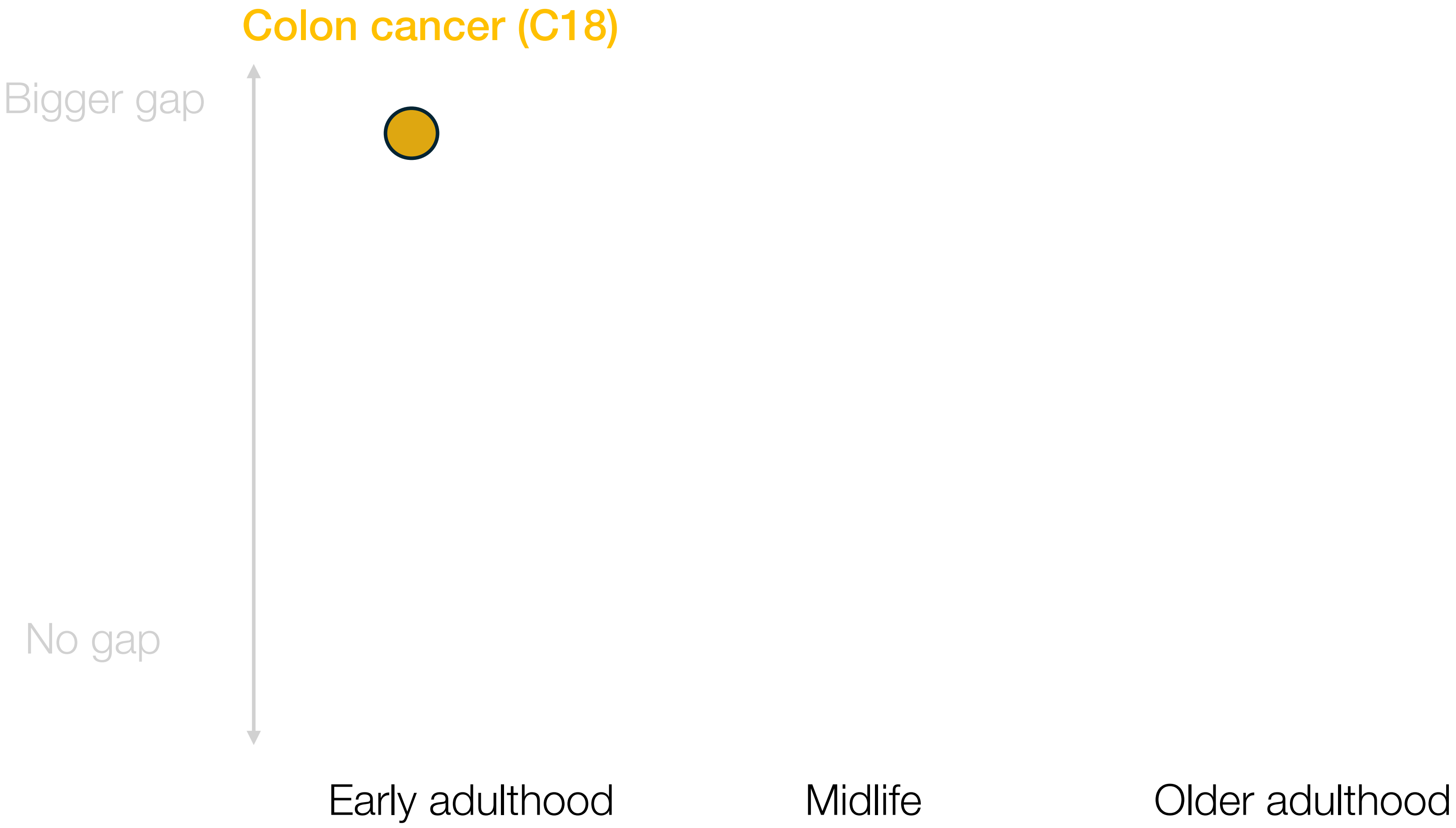
Compute for all diseases and see which reorganize the most:
F31 Bipolar disorder
T82 Mechanical complication
G44 Cluster headaches
E28 Ovarian dysfunction

Compare:

Does acquiring this condition reorganize the downstream disease landscape?

But when we split by age…





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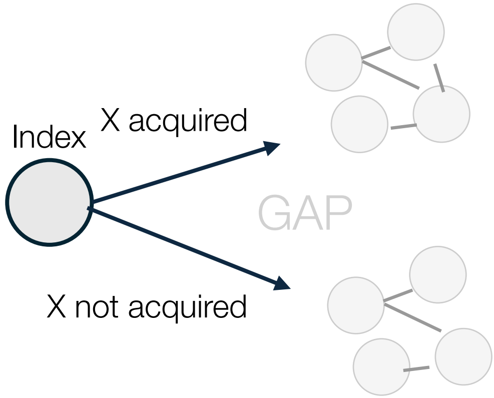

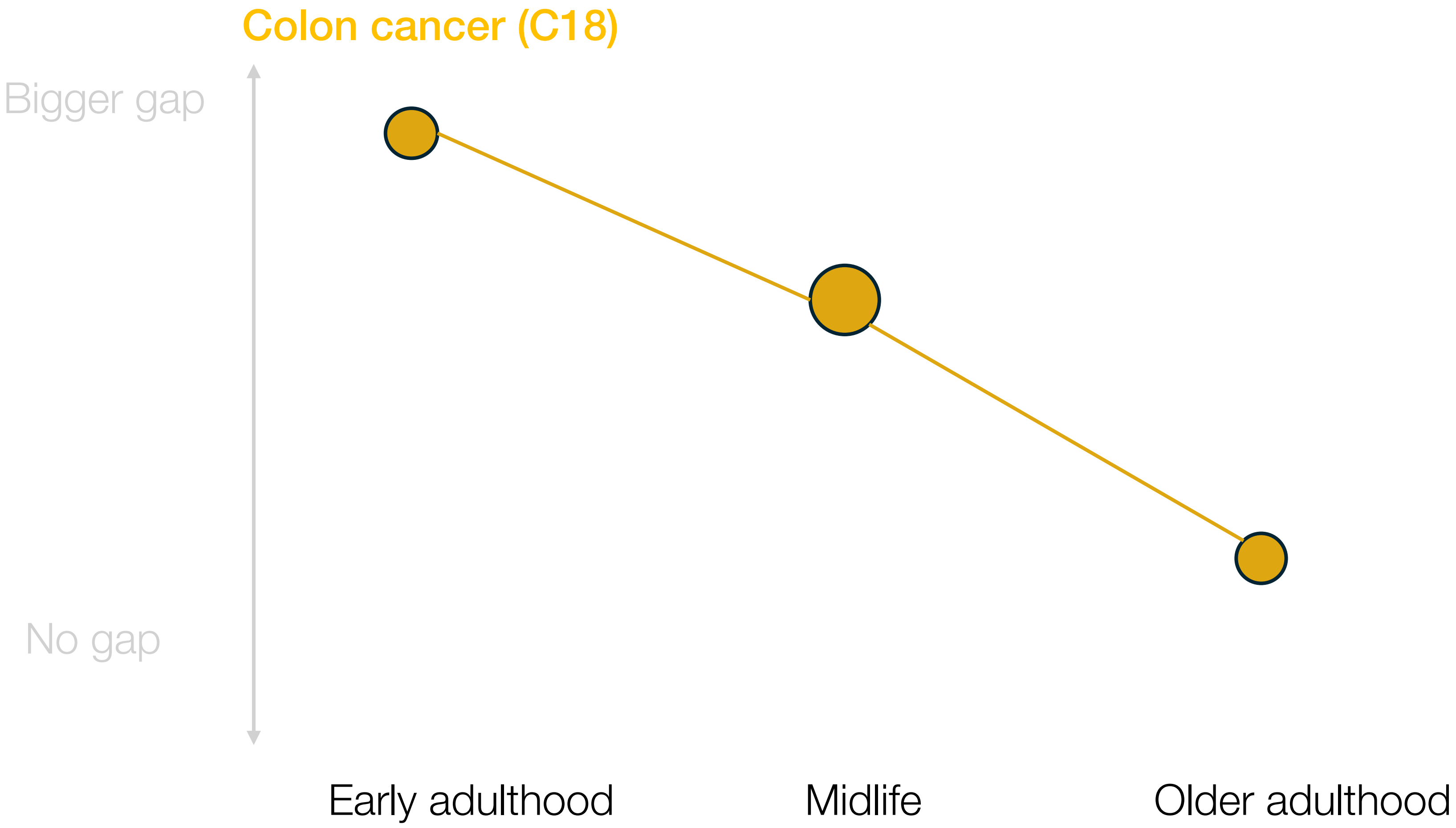

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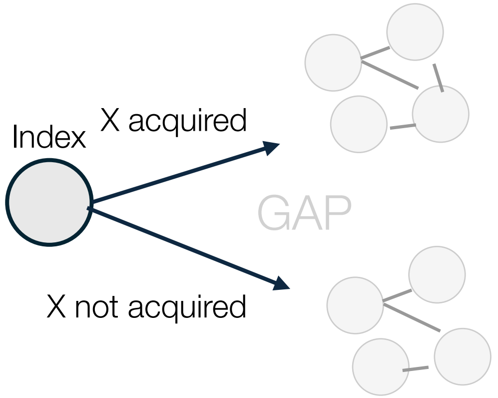

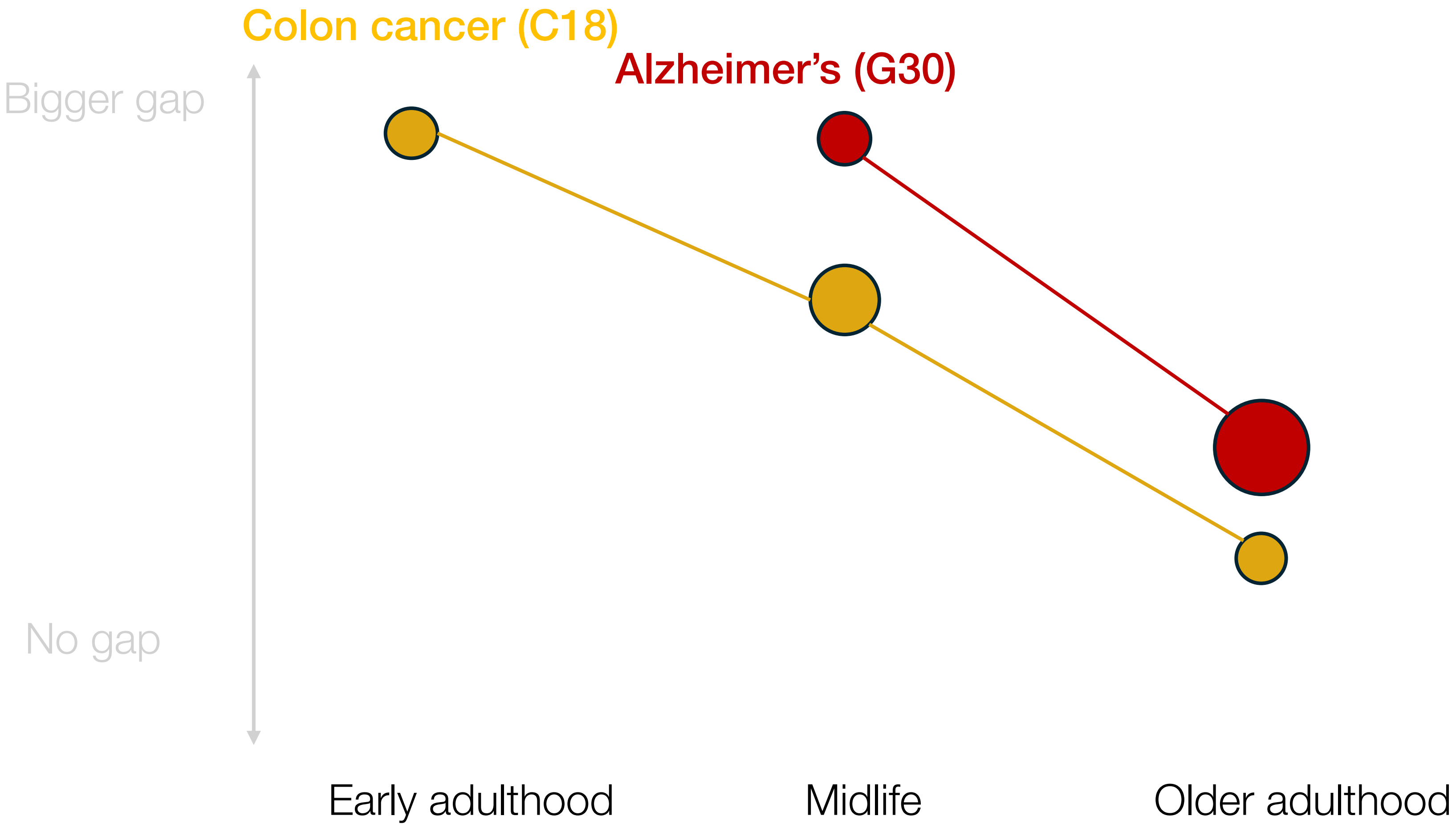

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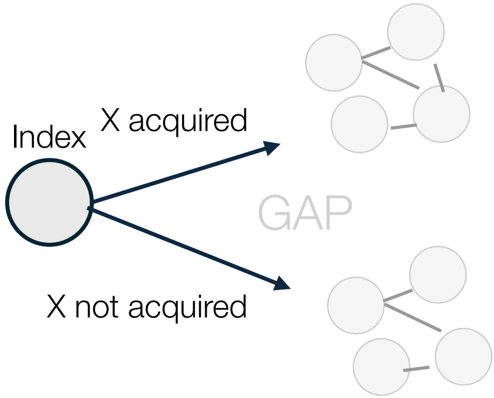

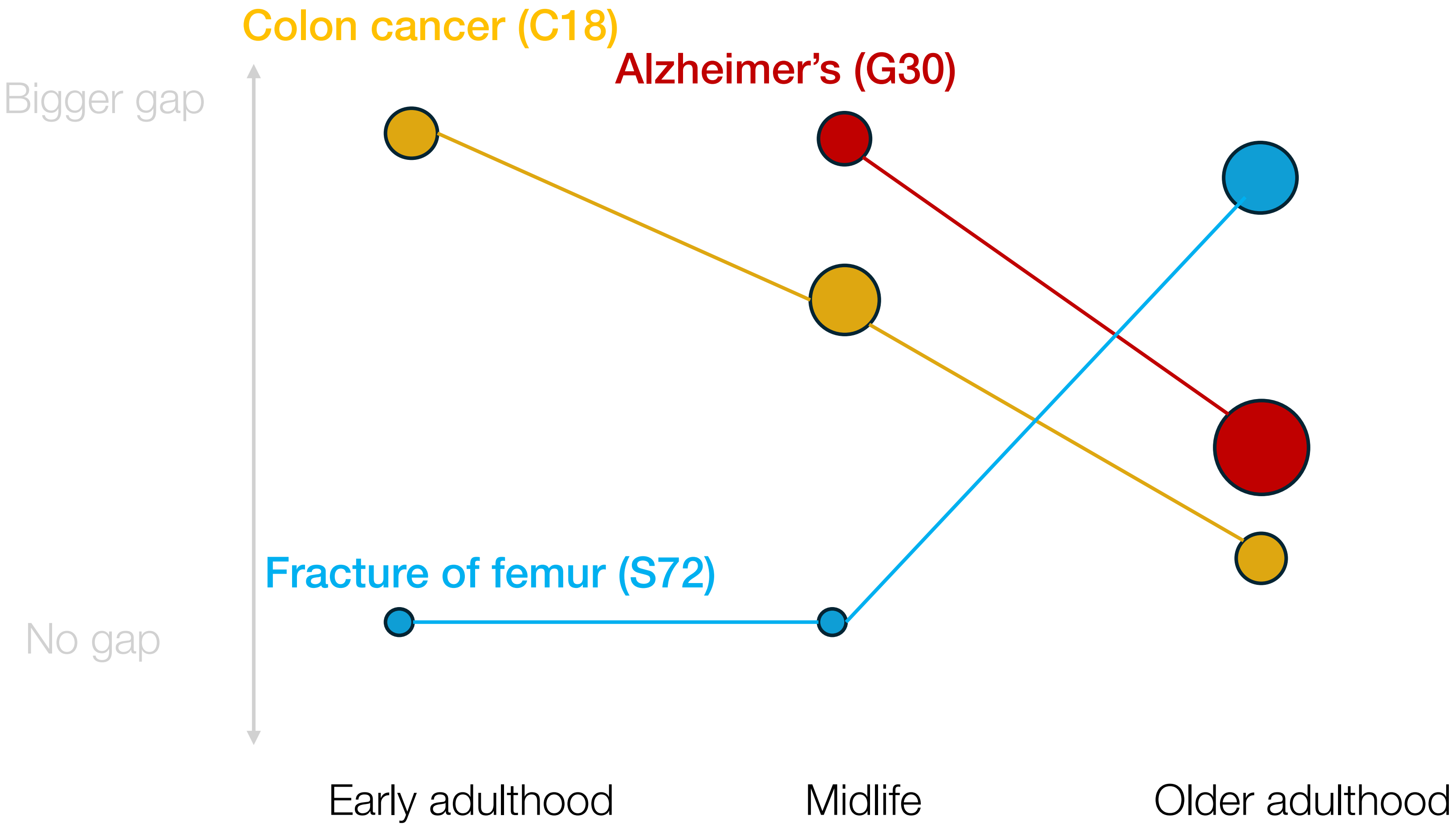

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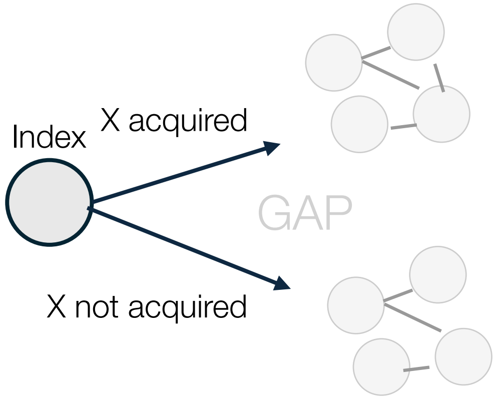

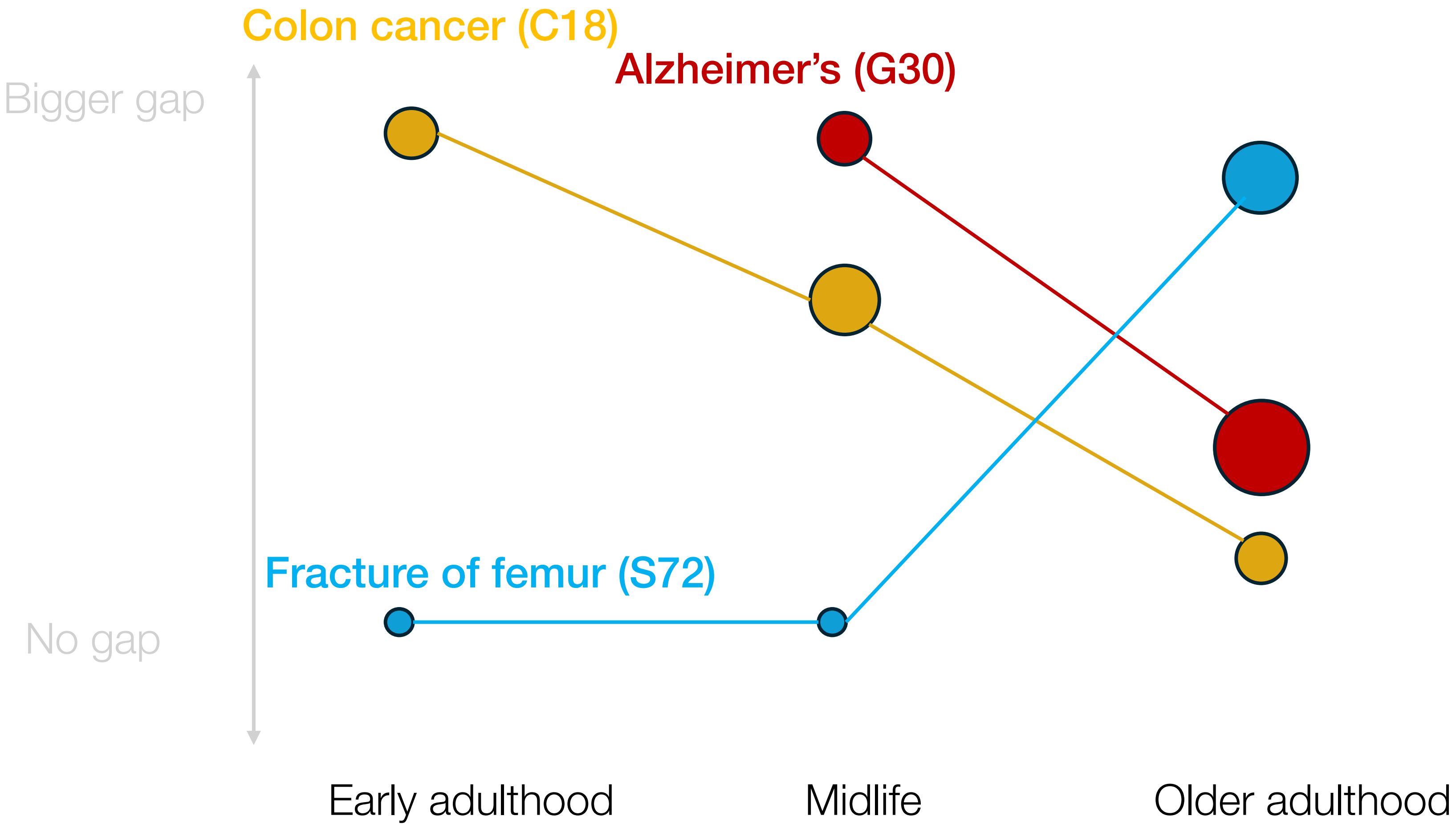

Summary

- U.S. primary care has a distinct disease trajectory landscape: chronic and episodic clusters with identifiable bridge nodes
- Since musculoskeletal conditions are upstream, treating them earlier may be important
- Bridge nodes and turning points are orthogonal: population-level structural centrality does not predict trajectory-redirecting effects
- A counterfactual design can detect conditions that persistently redirect the downstream disease landscape (turning points), which vary by age
- More validation needed to see if the method can help generate new hypotheses about disease relationships and intervention points

Thank you

David Rehkopf
Stanford

Nathaniel Hendrix
Amer. Board Fam. Med.

Diego Guerrero
UC Santa Barbara

Laust Mortensen
Univ. of Copenhagen

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So many thresholds to pick!

51% of pairs have fewer than 10 patients
76% have fewer than 50 patients
84% have fewer than 100 patients

n ≥ 100 + RR ≥ 1.5: 24,167 rows: across 10 bins that's ~2,400 edges per bin on average
n ≥ 500 + RR ≥ 1.5: 7,152 rows: ~715 edges per bin
n ≥ 1000 + RR ≥ 1.5: 3,830 rows: ~383 edges per bin

Inclusion criteria – full network

7,916,545 patients in system

5,774,940 patients have at least one visit (active)

3,586,684 patients have at least two visits, 6 months apart (62%)

The network has 10 time steps, one for each 6 month period.

Captures relative time, not absolute time.

Patients with visit span reaching END of each 6-month bin:

Bin	N patients	% of active
months 0-6	3,586,684	62.1%
months 6-12	3,070,024	53.2%
months 12-18	2,556,964	44.3%
months 18-24	2,145,565	37.2%
months 24-30	1,791,396	31.0%
months 30-36	1,504,426	26.1%
months 36-42	1,243,382	21.5%
months 42-48	1,023,085	17.7%
months 48-54	816,342	14.1%
months 54-60	625,973	10.8%

Community 0 (29 nodes):
Acute infectious disease
Bacterial infectious disease
Benign neoplasm of intra-abdominal organs
Bladder finding
Congenital disease
Disease caused by parasite
Disorder of endocrine system
Disorder of female genital system
Disorder of male genital organ
Disorder of pelvic region of trunk
Disorder of skin
Disorder of skin appendage
Disorder of the genitourinary system
Disorder of uterus
Evaluation finding
Female reproductive finding
Finding of abdominopelvic segment of trunk
Finding of pelvic region of trunk
Finding of substance level
Finding relating to sexuality and sexual activity
Inflammation of specific body systems
Inflammatory dermatosis
Lower urinary tract finding
Neoplasm of endocrine gland
Pain of truncal structure
Testicular finding
Urogenital finding
Vagina finding
Viral disease

Community 1 (5 nodes):
Abnormal finding on evaluation procedure
Visual system disorder
Neoplasm of intra-abdominal organs
Neoplasm of uterus
Primary malignant neoplasm of abdomen
Primary malignant neoplasm of trunk

Community 2 (12 nodes):
Anomaly of eye
Complication
Complication due to diabetes mellitus
Disorder due to type 2 diabetes mellitus
Disorder of carbohydrate metabolism
Disorder of eye
Infection by Ascomycetes
Measurement finding
Peripheral nerve disease
Retinal disorder
Retinopathy due to diabetes mellitus
Sequela of disorder

Community 3 (11 nodes):
Breast lump
Cognitive function finding
Cytopenia
Disorder of artery
Disorder of trunk
Finding of region of thorax
Hemoglobin low
Mass of body structure
Mass of trunk
Neoplasm affecting hematopoietic structure
Primary malignant neoplasm

Community 5 (15 nodes):
Abdominal organ finding
Abnormality of systemic vein
Colonic lesion
Disease due to Gram-positive bacteria
Disorder of anal canal
Disorder of colon
Disorder of digestive organ
Disorder of intestine
Disorder of large intestine
Disorder of pelvis
Disorder of the nose
Epithelial neoplasm of skin
High body weight
Injury of head
Tumor stage finding

Community 6 (9 nodes):
Disorder of gastrointestinal tract
Disorder of pregnancy
Fetal finding
Finding of measures of pregnancy
Finding related to pregnancy
Hypertensive disorder
Maternal AND/OR fetal condition affecting labor AND/OR delivery
Pelvic organ finding
Pregnancy, childbirth and puerperi um finding

Community 7 (2 nodes):
Congenital anomaly of head
Disorder of ocular adnexa

Community 8 (32 nodes):
Abnormal body temperature
Acute inflammatory disease
Acute respiratory disease
Allergic disorder
Bacterial respiratory infection
Complication of pregnancy, childbirth and/or puerperium
Degenerative disorder
Degenerative disorder of eye
Delivery finding
Disease due to Gram-positive coccus
Disorder due to infection
Disorder of face
Disorder of fetus or newborn
Disorder of lacrimal system
Disorder of upper digestive tract
Disorder of upper respiratory system
Finding of head region
Finding of mouth region
Functional finding of gastrointestinal tract
Head finding
Inflammation of specific body organs
Inflammatory disorder of head
Inflammatory disorder of upper respirator y tract
Intestinal infectious disease
Lesion of skin and/or skin-associated muc ous membrane
Pain in limb
Perinatal disorder of growth and/or develo pment
Poisoning
Respiratory tract infection
Upper respiratory infection
Viral respiratory infection
Vital signs finding

Community 9 (17 nodes):
- Arterial finding
- Chest injury
- Closed fracture
- Closed fracture of lower limb
- Closed fracture of upper limb
- Disorder of body wall
- Disorder of cardiac function
- Disorder of mediastinum
- Disorder of shoulder
- Disorder of thigh
- Disorder of thorax
- Disorder of vertebral column
- Fracture of bone
- Injury of free lower limb
- Injury of musculoskeletal system
- Injury of trunk
- Vascular disease of abdomen

Community 10 (33 nodes):
- Arthropod dermatosis
- Disorder of digit
- Disorder of extremity
- Disorder of finger
- Disorder of foot
- Disorder of free lower limb
- Disorder of hand
- Disorder of integument
- Disorder of joint region
- Disorder of lower extremity
- Disorder of soft tissue
- Disorder of soft tissue of head
- Disorder of soft tissue of limb
- Disorder of soft tissue of lower limb
- Disorder of upper extremity
- Finding of back
- Finding of hand region
- Finding of limb structure
- Finding of upper limb
- General clinical state finding
- Inflammation of specific body structures or tissue
- Injury of upper extremity
- Malignant neoplastic disease
- Musculoskeletal finding
- Neoplasm of skin
- Neoplastic disease
- Pain
- Skin lesion
- Soft tissue lesion
- Superficial injury
- Traumatic injury
- Traumatic or non-traumatic injury
- Wound

Community 11 (2 nodes):
- Disorder of lower respiratory system
- Disorder of lung

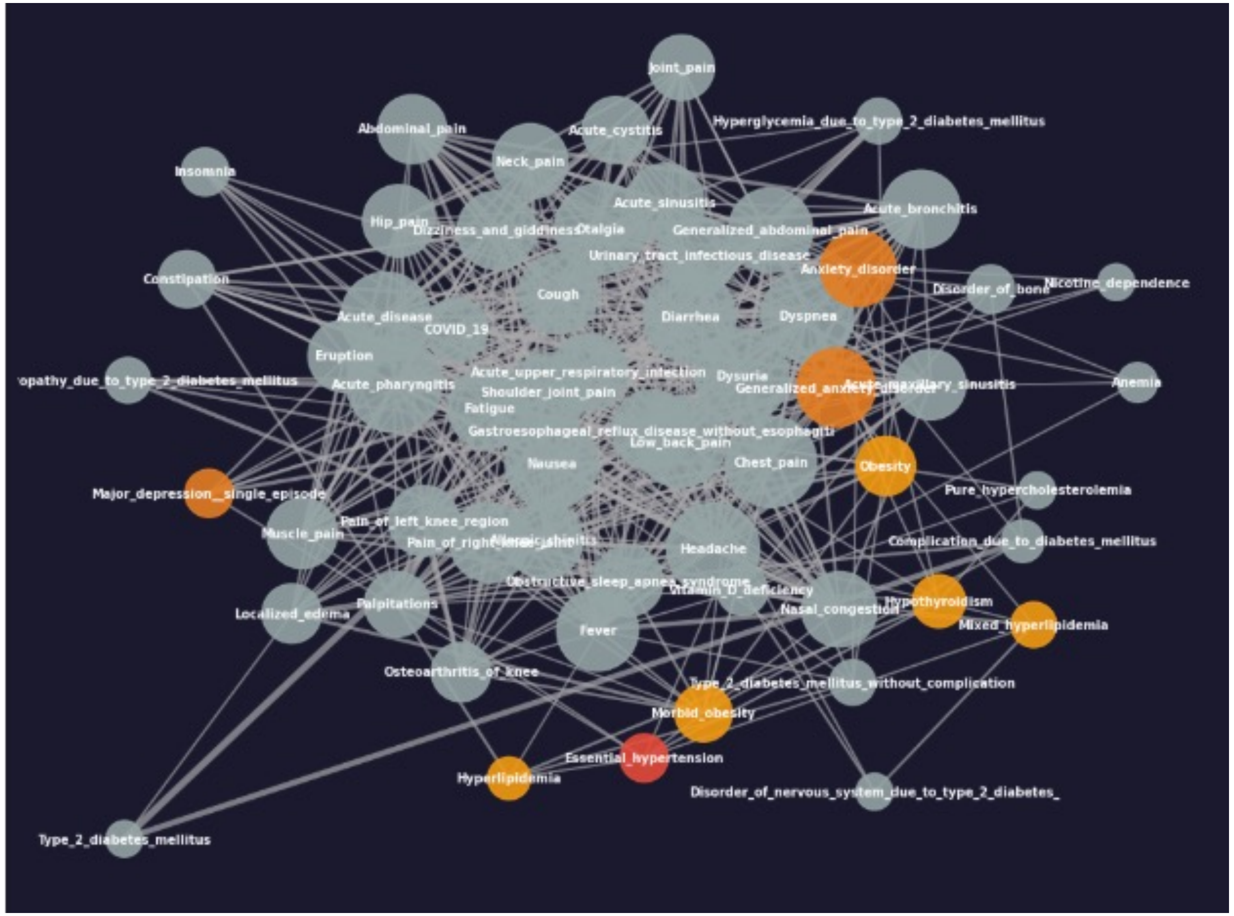
Community 12 (3 nodes):
- Blindness AND/OR vision impairment level
- Disorder of vision

Community 4 (101 nodes):

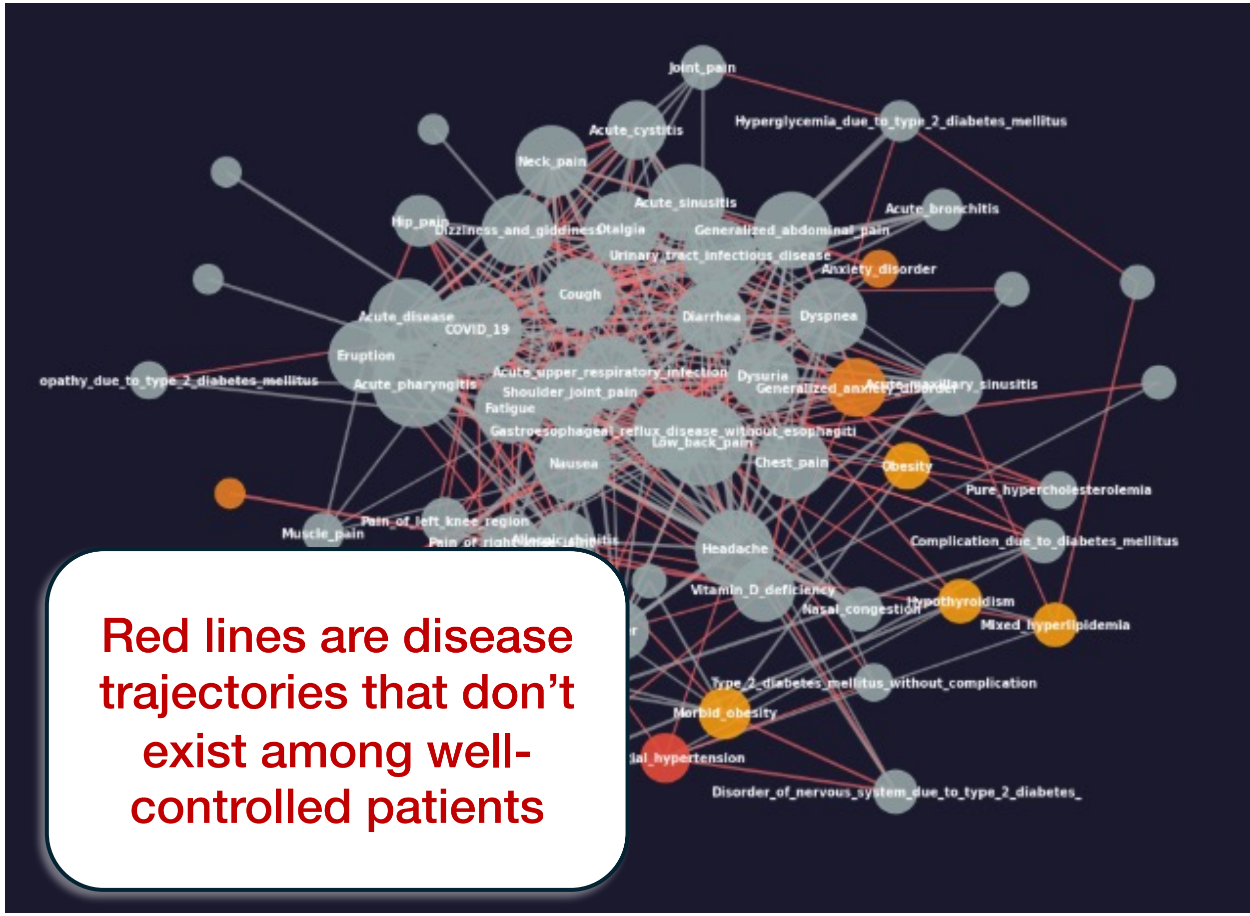
- Allergic disorder of respiratory tract
- Anemia
- Anxiety
- Arthropathy
- Atrial cardiopathy
- Behavior finding
- Bipolar disorder
- Cardiac arrhythmia
- Cardiac finding
- Cardiovascular finding
- Cardiovascular measurement - finding
- Chronic disease of respiratory system
- Deficiency of micronutrients
- Depressive disorder
- Developmental disorder
- Developmental mental disorder
- Disease due to Gram-negative bacteria
- Disease due to Paramyxoviridae
- Disease related state
- Disorder of abdomen
- Disorder of abdominopelvic segment of trunk
- Disorder of amino acid metabolism
- Disorder of auditory system
- Disorder of bone
- Disorder of brain
- Disorder of cardiac ventricle
- Disorder of cardiovascular system
- Disorder of cellular component of blood
- Disorder of connective tissue
- Disorder of ear
- Disorder of epiphysis
- Disorder of external ear
- Disorder of head
- Disorder of immune function
- Disorder of joint of spine
- Disorder of knee
- Disorder of lipoprotein AND/OR lipid metabolism
- Disorder of lipoprotein storage and metabolism
- Disorder of mineral metabolism
- Disorder of musculoskeletal system
- Disorder of neck
- Disorder of nervous system
- Disorder of prostate
- Disorder of psychological development
- Disorder of respiratory system
- Disorder of retroperitoneum
- Disorder of right ear
- Disorder of skeletal system
- Disorder of spinal region
- Disorder of stomach
- Disorder of the central nervous system
- Disturbance of consciousness
- Drug dependence
- Drug-related disorder
- Dyssomnia
- Ear, nose and throat disorder
- Emotional state finding
- Finding of functional performance and activity
- Finding of respiration
- Finding of sensation by site
- Finding of spinal region
- Finding of thought content
- Finding related to ability to move
- General finding of soft tissue
- Hearing disorder
- Heart block
- Heart disease
- Heart failure
- Hyperlipidemia
- Inflammatory disorder of extremity
- Inflammatory disorder of the respiratory tract
- Joint finding
- Kidney disease
- Knee joint finding
- Language finding
- Lesion of joint
- Level of consciousness - finding
- Measurement finding above reference range
- Mental disorder
- Mental state finding
- Mental state, behavior and/or psychosocial function finding
- Metabolic disease
- Mood disorder
- Musculoskeletal and connective tissue disorder
- Mycosis
- Necrosis of anatomical site
- Neurological lesion
- Pain / sensation finding
- Respiratory finding
- Seizure disorder
- Sensory nervous system finding
- Sequela
- Speech finding
- Sprain of joint
- Structural disorder of heart
- Tracheobronchial disorder
- Traumatic or nontraumatic brain injury
- Undernutrition
- Urinary system finding
- Vascular disorder
- Viscus structure finding

The disease landscape is **structurally different** between groups

Well controlled diabetes (Hba1c < 7)



Poorly controlled diabetes (Hba1c > 9)

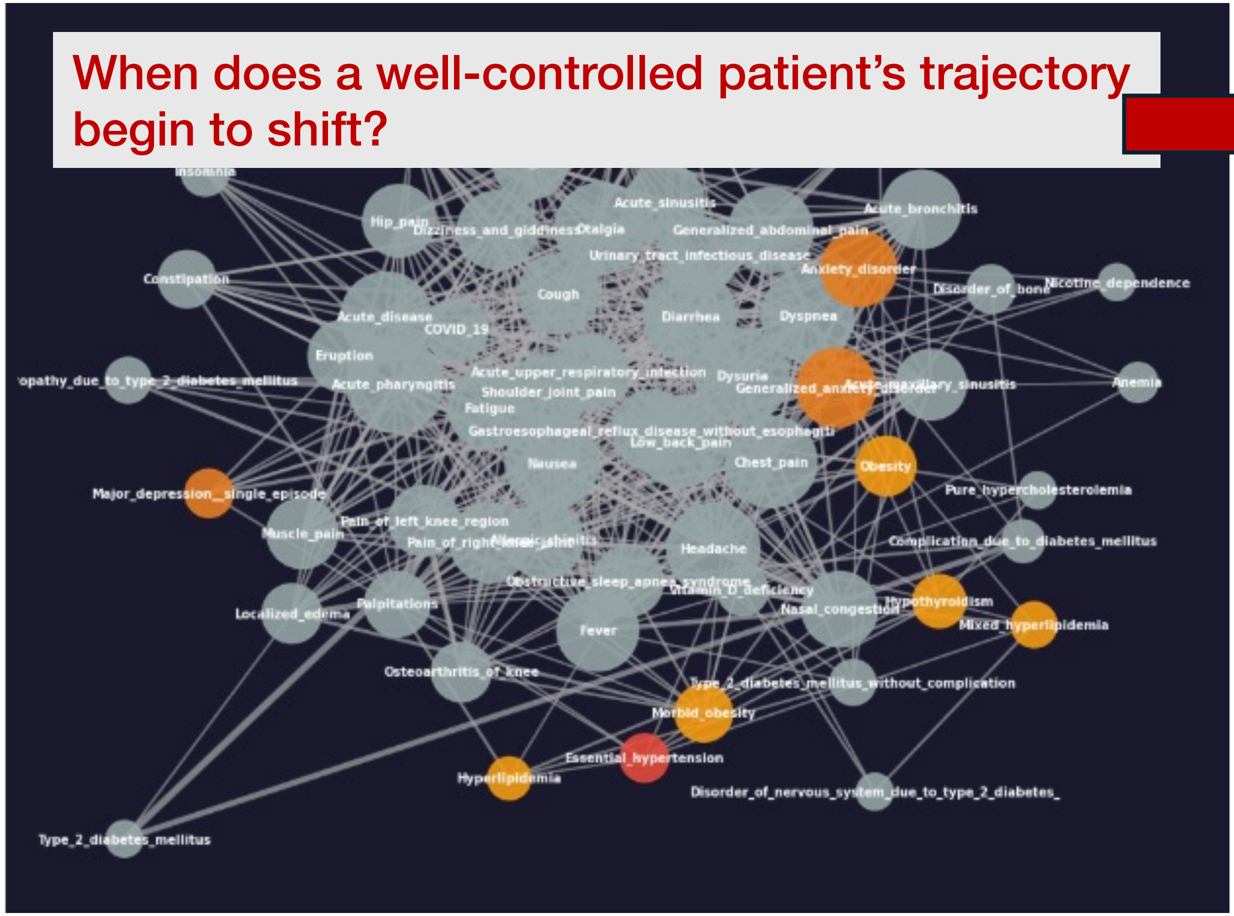


Red lines are disease trajectories that don’t exist among well-controlled patients

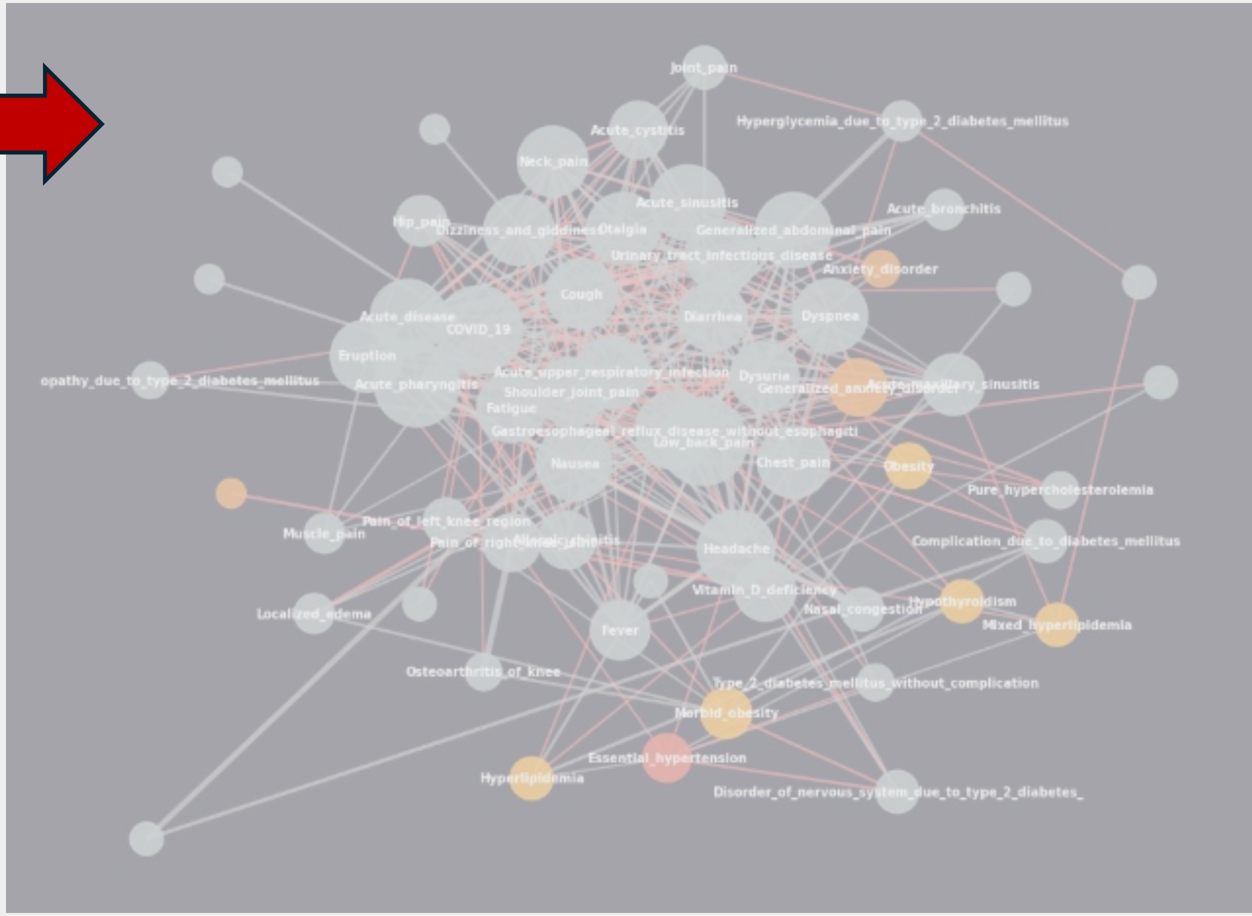
The disease landscape is **structurally different** between groups

Well controlled diabetes (Hba1c < 7)

When does a well-controlled patient's trajectory begin to shift?

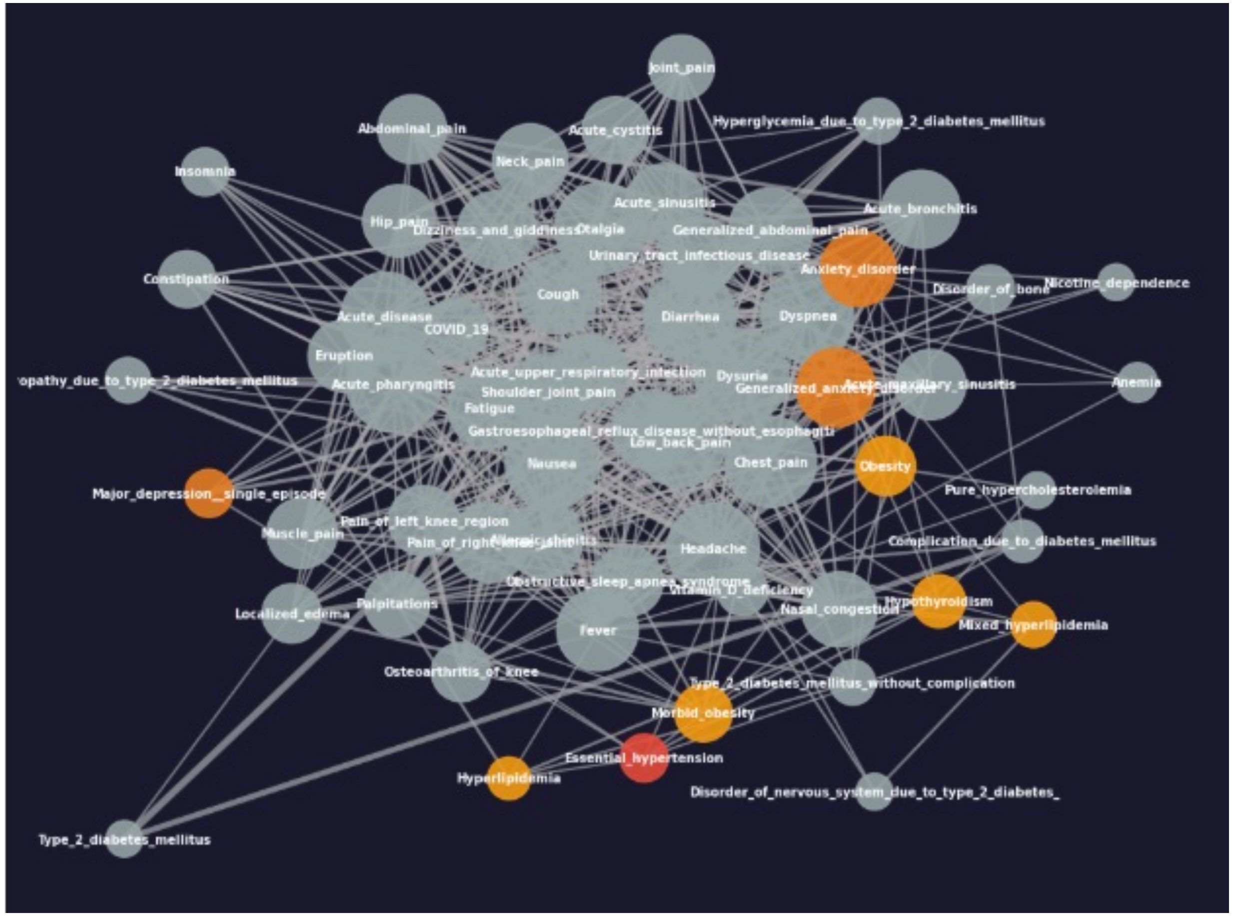


Poorly controlled diabetes (Hba1c > 9)



Depression is a possible turning point

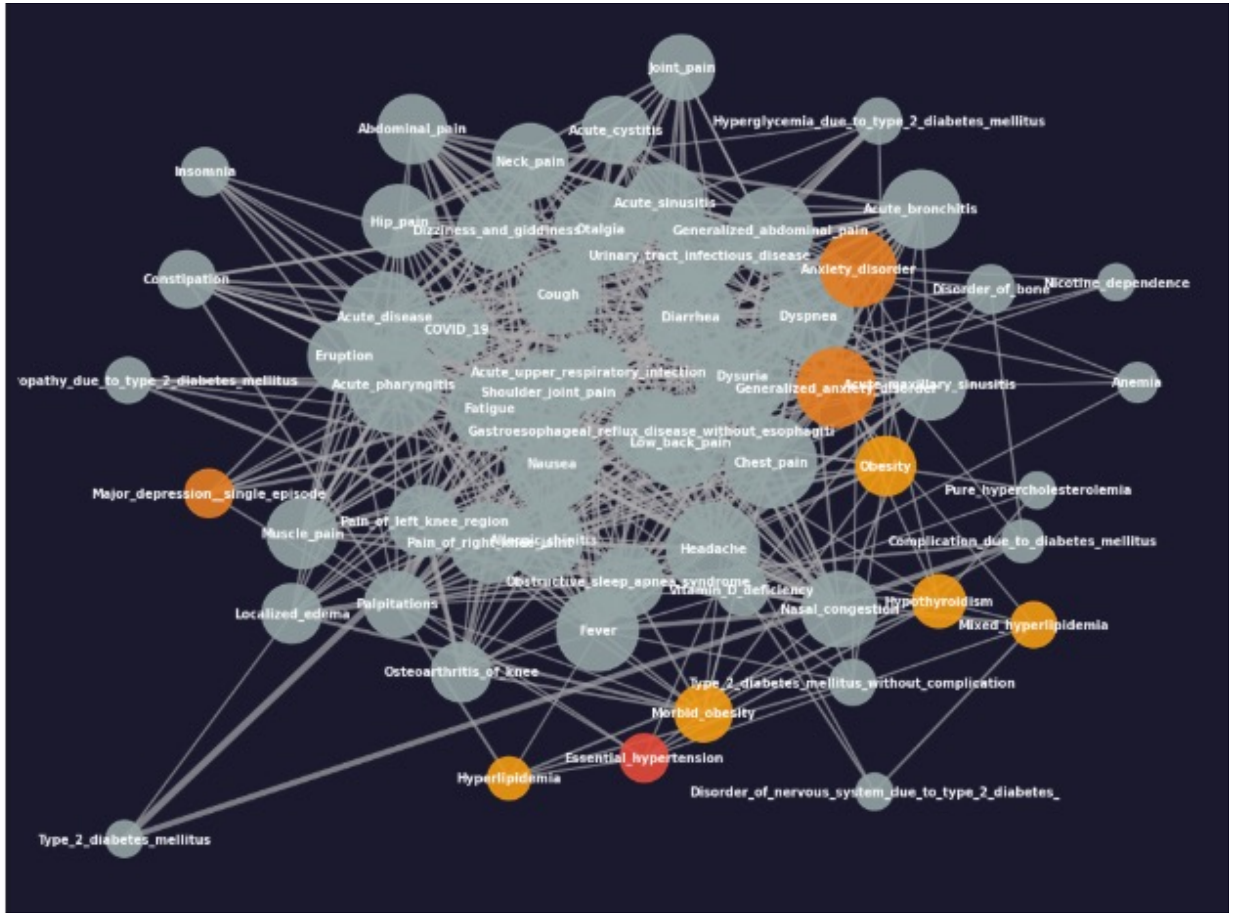
Well controlled diabetes (Hba1c < 7)



Well-controlled patients diagnosed with **depression** have **more chronic conditions** later, than those not diagnosed with depression

Depression is a possible turning point

Well controlled diabetes (Hba1c < 7)



Well-controlled patients diagnosed with depression have **more chronic conditions** later, than those not diagnosed with depression

If we can identify turning points, we can intervene before complications compound

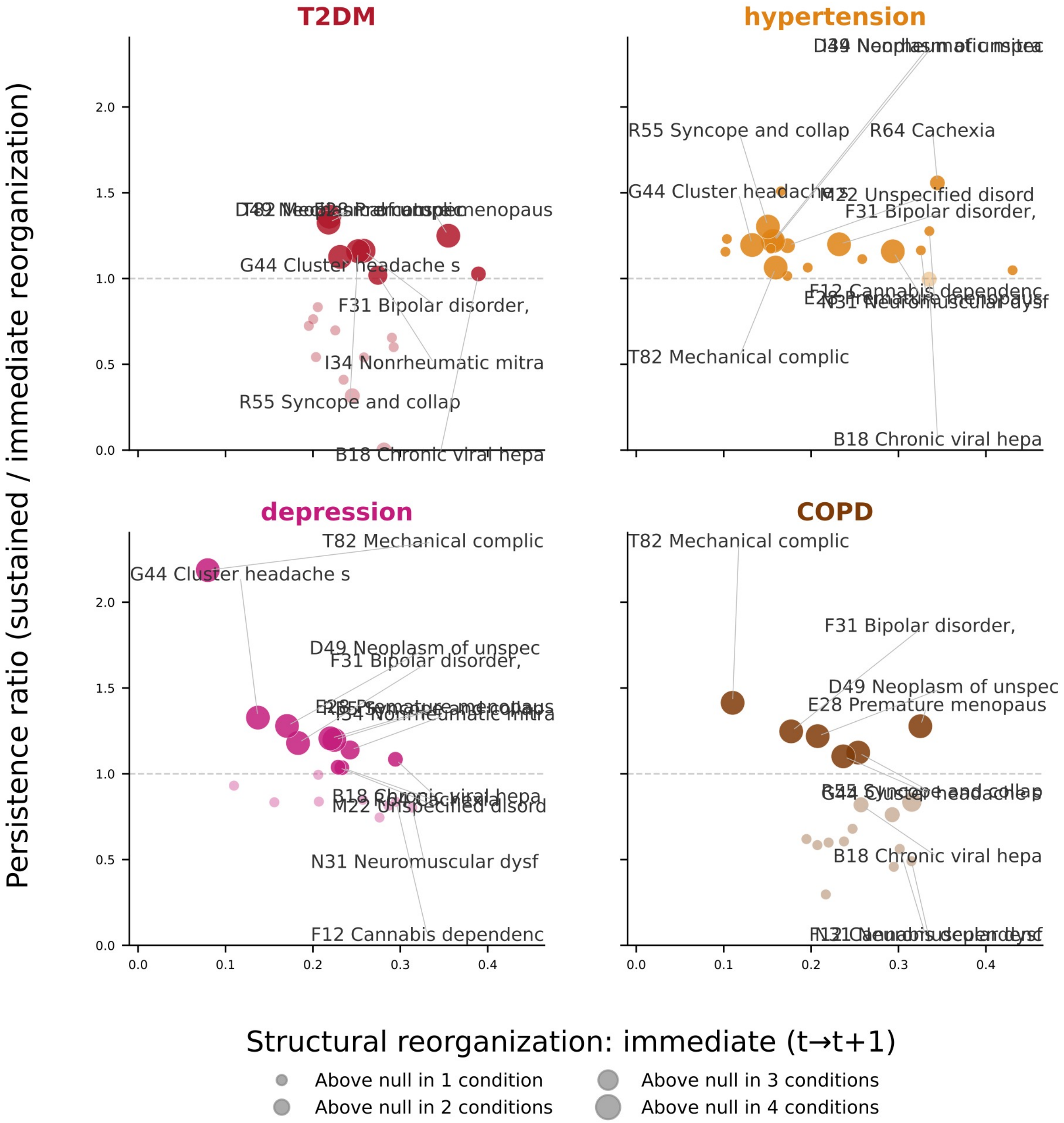
Turning points

A condition is a turning point if acquiring it:

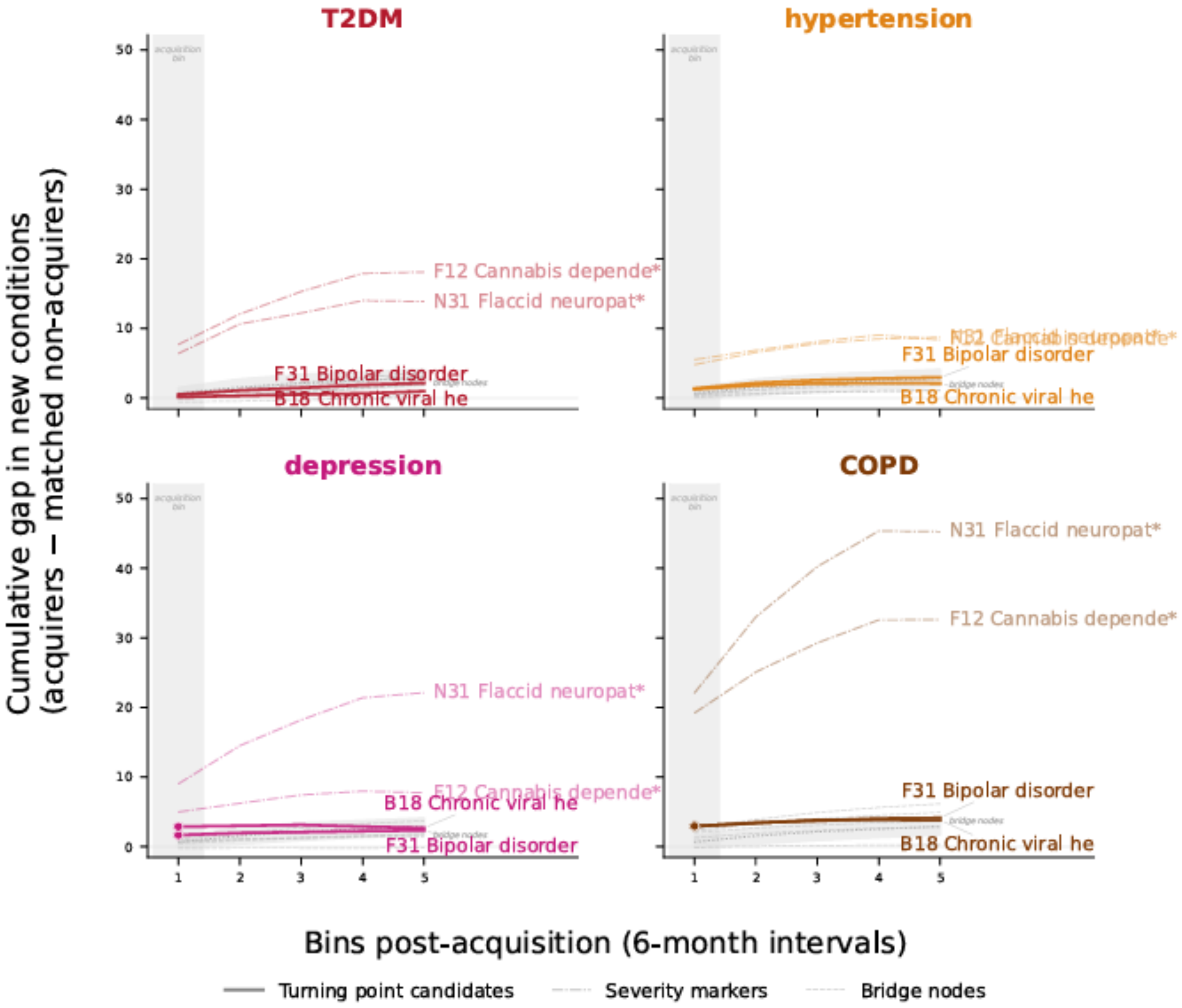
1. Restructures the downstream disease network (edge rewiring)
2. Does so **persistently** across time

Tested via matched acquirer / non-acquirer comparison

Does that effect strengthen or attenuate over time?

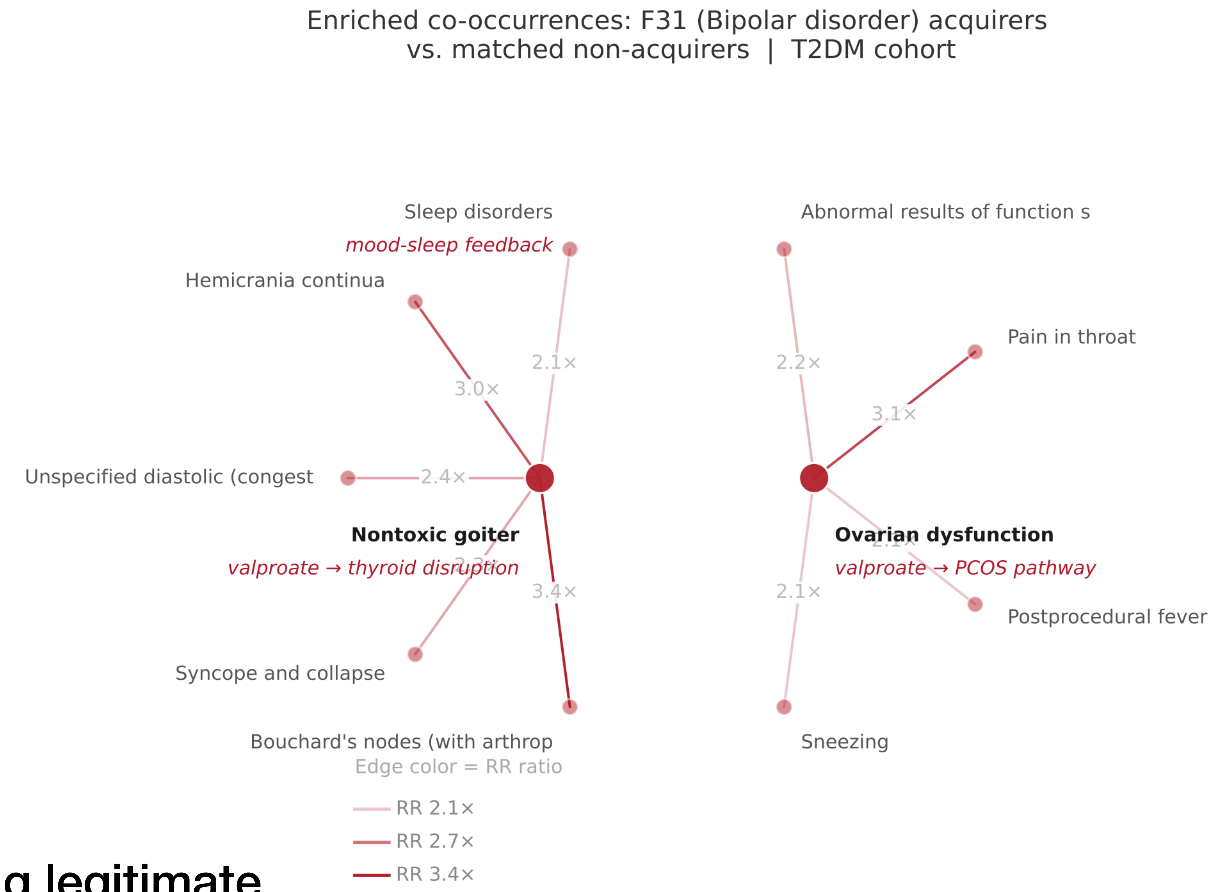


To what degree does a condition rewire the network? (% edges changed)


T2DM
hypertension
depression
COPD
Cumulative gap in new conditions
(acquirers − matched non-acquirers)
Bins post-acquisition (6-month intervals)
F12 Cannabis depende*
N31 Flaccid neuropat*
F31 Bipolar disorder
B18 Chronic viral he
Turning point candidates
Severity markers
Bridge nodes

Recovering a known turning point

All of these connections exist only among those who acquired bipolar disorder, which opened a whole new part of the network



It seems like it is recovering legitimate turning points (known from valproate literature)

Bridge nodes ≠ turning points

- Bridge nodes (high betweenness centrality in the population network) do not appear as turning point candidates

- Structural centrality in the population-level network and trajectory-redirecting effects at the individual level are orthogonal

- Turning points must be detected dynamically: static network structure is not sufficient